

Chaos based image encryption scheme to secure sensitive multimedia content in cloud storage

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ABSTRACT

Cloud computing offers a variety of on-demand services to users and has gained significant prominence in the contemporary era. The security of information stored in cloud data centers has become a central concern, especially for sensitive data like medical images, videos, and multimedia content that require heightened protection when stored in data centers. Cloud users have a responsibility to ensure the security of their data through strategic measures. Focusing on maintaining the privacy of images stored in the cloud, this research introduces an innovative image encryption technique that is based on a modified Skew Tent chaotic map. The suggested modification to the chaotic map includes combining the Skew Tent map with both the sine trigonometric function and perturbation technology. This results in enhanced randomness, more intricate dynamical behavior, a broader chaotic range, and increased sensitivity to initial values in contrast to alternative chaotic maps. The incorporation of this adapted map into a stepwise procedure, involving two rounds of permutation followed by diffusion, effectively accomplishes image data encryption for cloud storage systems. These consecutive operations collectively enhance the encryption method's randomness and robustness. Through simulations conducted using Cloudsim, the cipher-image exhibits a uniform distribution and achieves a commendable information entropy score of 7.996749. The encryption algorithm significantly reduces correlation coefficients from 1 in the original image to 0 in the encrypted image, while maintaining NPCR (Number of Pixel Change Rate) and UACI (Unified Average Changing Intensity) values within the critical range. Additionally, both theoretical analysis and practical evaluations confirm the algorithm's resilience against exhaustive, occlusion, and classical attacks. Moreover, extending this encryption framework to video data, a novel video encryption approach is proposed.

1. Introduction

Cloud computing is a significant technical advancement in the information era and knowledge economy of the twenty-first century. It is a major technological breakthrough that makes data, apps, and services accessible to users anywhere in the world at any time via the internet. It simultaneously reduces data storage costs, relieves storage platform burdens, and offers scalability, dependability, and improved data processing capabilities (Buyya et al., 2009; Li, Yu, Song et al., 2021). Cloud computing is very popular in the IT industry because it is scalable, cost-effective, easy-to-use, collaborative, and offers advanced services. It also allows for quick adoption and innovation, as well as high performance and a global infrastructure. The forecast from Gartner, a leading research and advisory company, says that public cloud services will be worth more than \$1 trillion by 2026. This is a very fast rate of growth that is faster than any other IT market (Gartner, 2022). In cloud computing, there are generally three main service types

available: “public, private, and hybrid clouds”. The public cloud is acquiring popularity among businesses and individuals as a result of its increased flexibility, operational cost savings, and enhanced customer support, resulting in a rise in its adoption (Zhang et al., 2021). Indeed, the security of data stored in a cloud environment continues to be an important concern that requires meticulous care. Academics and researchers are conducting extensive research on cloud-based data security and encryption. Fig. 1 shows that the number of scholarly publications on cloud data security and encryption methods is going up. The abbreviations used throughout this manuscript are summarized in Table 1.

1.1. Motivation

Cloud-based data sharing provides user-friendly advantages over alternative methods. The past decade has witnessed the rise of multimedia technology and intelligent devices, which have facilitated the

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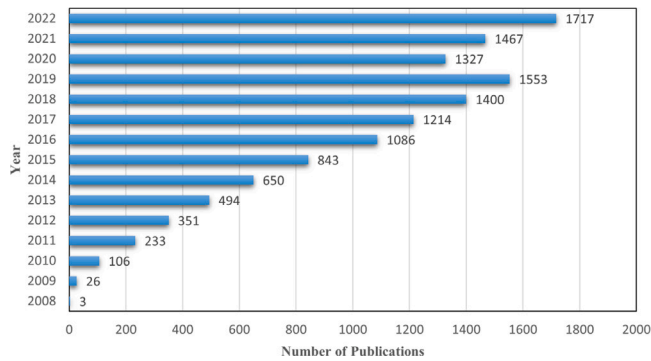


Fig. 1. The count of scholarly publications on data security and encryption in a cloud environment since 2008.

Table 1

List of abbreviation used in the paper.

S. No	Abbreviation	Full form
1	AES	Advanced Encryption Standard
2	DES	Data Encryption Standard
3	C-IES	Chaos based Image Encryption Scheme
4	STM	Skew Tent Map
5	S-STM	Sine Skew Tent Map
6	LE	Lyapunov Exponent
7	FrDCT	Fractional Discrete Cosine Transform
8	CA	Cellular Automata
9	IIoT	Industrial Internet of Things
10	PLSM	Piecewise Logistic-Sine Map
11	PRNS	Pseudo Random Number Sequence
12	RNS	Random Number Sequence
13	PRNG	Pseudo Random Number Generator
14	PWLCM	Piecewise Linear Chaotic Map
15	DNN	Deep Neural Network
16	CRT	Chinese Remainder Theorem
17	CS	Compressive Sensing
18	LTSS	Logistic-Tent-Sine Chaotic Map
19	DNA	DeoxyriboNucleic Acid
20	MLE	Maximum Lyapunov Exponent
21	SSTM-IES	S-STM Based Image Encryption Scheme
22	SHA3-256	Secure Hash Algorithm 3, 256-bit
23	MSE	Mean Squared Error
24	PSNR	Peak Signal to Noise Ratio
25	NPCR	Number of Pixel Change Rate
26	UACI	Unified Average Changing Intensity
26	SSIM	Structural Similarity Index Measure

accessibility of digital image data (Li, Yu, Song et al., 2021). Digitalized sharing via contemporary devices has revolutionized the global distribution of information across various industries, such as telemedicine, education, and biometric authentication, which rely heavily on image context (Dagadu et al., 2019; Lakshmi et al., 2021). However, in practise, security is often overlooked. This is because cloud computing operates in a public setting managed by untrusted third parties, placing user data at risk. In order to safeguard private and confidential image data from unauthorized access and use in a variety of industries, stringent security measures such as encryption, access controls, and secure storage are required (Vengadapurvaja et al., 2017). To keep picture data safe, it is important to put in place strong security measures like encryption, access controls, and safe storage. The security of image files is of utmost importance to avoid the accidental or malicious release or use of private or sensitive information. Traditional encryption algorithms like DES or AES have trouble dealing with the significant correlation of images, hence newer encryption technologies like chaotic image encryption techniques have gained favor to maintain image security (Lai et al., 2022; Su et al., 2023).

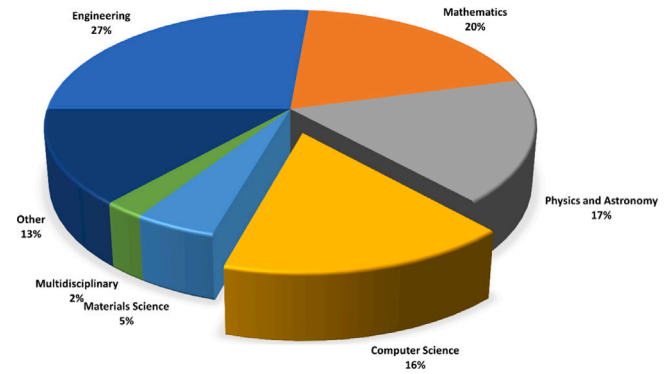


Fig. 2. Publications on chaos theory on the basis of subject area.

1.2. Problem definition

Chaos theory holds significance across various disciplines, including computer science, engineering, sociology, physics, economics, biology, and philosophy, mathematics, as evidenced by the extensive range of research papers (see Fig. 2) (Lambić & Nikolić, 2017). Chaotic systems have grown in prominence in cryptography due to their sensitivity to beginning conditions, non-periodicity, and randomness (Li, Yu, Wang, 2021). These systems can be classified as either one-dimensional or high-dimensional chaotic maps. One-dimensional chaotic maps are ideal for low-power devices due to their simple structure and efficient iteration. However, they are restricted in the range of parameters they can handle and the intricacy of their dynamic behavior. As a result, they are susceptible to attacks that attempt to predict erratic sequences, posing security risks for practical applications. However, high-dimensional chaotic maps are exceptionally hard to forecast due to their complex structures and unpredictable behaviors. But putting these maps into hardware takes a lot of computing power, which raises questions about efficiency (Zhang et al., 2020). Therefore, it is crucial in the design of chaos-based encryption systems to create chaotic maps that provide optimal tradeoffs between efficiency and security.

In order to enhance digital chaos and broaden the parameter range of digital chaotic maps, a variety of modified chaotic models have been developed. Certain research focuses specifically on increasing the complexity of chaotic behaviors (Alawida et al., 2019). Additionally, a variety of chaos-based encryption techniques have been developed. Nonetheless, certain Chaos-based Image Encryption Schemes (C-IESs) have experienced problems. Numerous adversaries have analyzed and outlined the security flaws of these algorithms (Abanda & Tiedeu, 2016; Preishuber et al., 2018). This research offers the Sine Skew Tent Map (S-STM) as a remedy to the skew tent map's shortcomings. Furthermore, we suggest an S-STM-based cloud-based picture encryption storage system. The encryption system was created with the goal of enhancing aspects related to cryptography while maintaining a balanced approach between the chaotic map's efficiency and complexity.

1.3. Research gaps

The majority of the current main solutions for color image encryption are chaotic systems, which are characterized by their inherent properties of pseudo-randomness and sensitivity to initial conditions. These qualities bolster security by employing intricate permutation and diffusion mechanisms. Some noteworthy approaches are the use of hybrid techniques combining multiple chaotic maps, content-sensitive dynamic function switching mechanisms, and 3-D chaotic maps paired with shuffling algorithms (Zhou et al., 2023).

However, there are still notable areas of research that have not been adequately addressed. Current techniques necessitate significant

refinement in order to improve their resilience against advanced assaults, such as chosen ciphertext and adaptive chosen plaintext attacks. Furthermore, there is still a lack of resilience against a variety of cropping and noise assaults (Hua et al., 2021; Midoun et al., 2021).

To improve time efficiency — which is vital for cloud data applications — it is imperative to guarantee good security with low computational complexity, especially in environments with parallel processing. It is still difficult to balance effective encryption speed with strong security, particularly for real-time applications. Additional optimization is necessary to decrease computational complexity while preserving strong encryption. It is crucial to address these deficiencies in order to create picture encryption solutions that are more effective, safe, and feasible.

1.4. Proposed study

This work presents a new image encryption algorithm designed to improve the security of data saved in cloud computing environments. The proposed solution overcomes the constraints of the original Skew Tent Map (STM) by introducing the Sine Skew Tent Map (S-STM). This adaptation eliminates trap situations and enhances sensitivity to initial conditions. The enhanced chaotic qualities of the S-STM, such as its wider key space, periodicity, and increased unpredictability, make it a strong option for picture encryption in cloud storage systems.

The encryption procedure consists of two successive rounds of permutation, which are then followed by diffusion. This process greatly improves the strength and unpredictability of the encryption. In an effort to enhance security, hash-based keys are integrated, offering defense against chosen-plaintext and known-plaintext attacks. The suggested encryption technique has undergone comprehensive statistical and security evaluations on a diverse database, showcasing strong resilience against typical attacks. In addition, the study expands the encryption framework to include video data and introduces a new video encryption technique that is also assessed for its effectiveness. The primary contributions of this study may be summed up as follows:

- The Sine Skew Tent Map (S-STM) is adapted from the Skew Tent Map (STM) to solve its shortcomings, such as the elimination of trap circumstances and the limited sensitivity to beginning conditions shown in the original STM.
- The S-STM possesses enhanced chaotic properties, including increased unpredictability, periodicity, and a larger key space.
- The S-STM has been used as a core component of a proposed image encryption scheme for cloud storage.
- To increase the algorithm's security, hash-based keys are incorporated to protect against chosen-plaintext and known-plaintext attacks.
- The two round permutation method employed to concurrently execute the confusion operation and enhance the encryption process's resilience.
- The proposed encryption technique was thoroughly examined and discovered to be very resistant to common attacks, as evidenced by statistical and security assessments on a diversified database.
- Using the same Image encryption strategy, a video encryption method is also suggested here. To illustrate the efficacy of this methodology, a comparative examination is provided.

This research not only contributes to the progress of chaotic image encryption but also offers a practical and strong approach for improving data security in cloud environments. The suggested encryption solution based on S-STM greatly enhances the security of sensitive data, effectively tackling key difficulties in cloud computing. The method ensures a high degree of unpredictability and resilience against common assaults, which helps to preserve user trust and safeguard sensitive information across a wide range of applications, such as medical, educational, and biometric industries. The expansion of the encryption technique to video data further showcases its adaptability and potential for other applications, rendering it a comprehensive solution for safeguarding multimedia content in cloud storage systems.

1.5. Organizational structure of the paper

The remainder of this paper's organizational structure is as follows: In Section 2, we will review relevant literature concerning the application of chaotic maps in the development of image encryption schemes. Section 3 will explore the Skew Tent Map and the challenges associated with it, while Section 4 will present the innovative Sine Skew Tent Map (S-STM). This section will do a comparison study, looking at the chaotic range and Lyapunov exponent (LE) values of other current chaotic maps in addition to the S-STM for a thorough evaluation. Section 5 will outline the proposed image encryption algorithm, and Section 6 will detail the decryption process. Section 7 will present the outcomes and analysis of the image encryption system. In Section 8, a new video encryption method will be introduced, accompanied by a security analysis. Section 9 will discuss the superiority, applicability, and limitations of the proposed encryption scheme. The paper concludes with Section 10.

2. Related work

This section presents an overview of the current literature concerning the application of chaos-based cryptography in encryption algorithms. A significant portion of the studies has focused on incorporating chaotic system in diverse domains, such as pseudorandom number generation, audio encryption, image encryption, and the integration of encryption within embedded systems.

In 2019, Hu et al. (2020) developed a safe image cryptosystem by fusing matrix convolution, a chaotic Fibonacci system, and a cloud computing model. They started by breaking down a color picture into its red, green, and blue components. They generated a chaotic sequence to permute pixel points and perform matrix convolution using the Fibonacci chaotic system. To achieve the shuffling of pixel locations, this chaotic sequence was applied as a rectangular matrix to all three components: Red, Green, and Blue. The result was then convoluted with a matrix produced by a Fibonacci chaotic system modeled by clouds. The intermediate stage was then XORed with a collection of previously-generated chaotic sequences. Finally, the rectangular image was transformed into a color image that had been encrypted. Kumar et al. (2019) presented a novel cryptographic technique designed specifically for medical information. The encryption process involved implementing the fractional discrete cosine transform (FrDCT) on the original image and subsequently applying a chaotic map to the obtained FrDCT coefficients. To validate the effectiveness of their proposed technique, the researchers employed both theoretical analysis and practical experiments. The results indicated that the scheme offers improved adaptability for encrypting medical images while maintaining robustness against a variety of attacks, such as differential, statistical, and brute-force attacks. Mondal et al. (2019) investigated extremely secure picture encryption algorithms for safe image communication and storage. Their technique is based on a sequence generated by a chaotic map STM and cellular automata (CA). The generated sequence serves as the CA's first vector generator. The CA requires a 128-bit starting bit sequence in order to build a PRNS. The resulting RNS is then used to permute the picture in order to minimize correlations between neighboring pixels. The picture also passes through encryption using a single random number and the RNS. It is clear from the suggested approach that combining a chaotic map with CA may improve the key space and speed up the RNS's creation.

In 2020, Zhang (2020) a straightforward image encryption algorithm based on the lifting scheme is proposed. The algorithm utilizes "transformation, permutation, and XOR diffusion" for its cryptographic scheme. Transformation is achieved by applying the image's discrete wavelet transform. Permutation is carried out through the use of S-box substitution. In order to accomplish diffusion, the technique uses a keystream generated by a simple pseudo-random number generator

(PRNG) that depends on the PWLCM chaotic map to shuffle the image's pixels. However, it is worth noting that the chosen substitution method, using the AES S-box, is not considered secure due to its lack of dynamic properties. [Falmari and Brindha \(2020\)](#) designed an authentication system using Paillier encryption, where a difference function and Homomorphic cryptosystem were employed. Their proposed image encryption technique draws inspiration from the Fridrich architecture and employs chaos-based keystreams. The cryptosystem ensures confusion at the bit level and diffusion at the pixel level. To enhance security, the keys are modified in each round of encryption, introducing further complexity to the system. [Maniyath and V \(2020\)](#) came up with an effective way to encrypt images using deep neural networks and chaotic systems. This improved the chaotic mapping and led to better key generation. The first step of the method is to take a picture as input and use random values to create an identically sized random matrix. A DNN network is used to train this matrix. The output that is produced next is used to create the secret key during both forward and backward transmission. The image's secrecy is then preserved by using this key.

In 2021, Using 2-D Moore Cellular Automata, [Roy et al. \(2021\)](#) created a realistic and efficient method for encrypting photos. This method was created especially for IoT devices with constrained resources. It uses local transformations that are based on the binary state values of nearby bits to make cellular automata implementation simpler and use fewer resources. The researchers looked at two different iterations of the algorithm, one using a null boundary and the other using a periodic boundary for the neighborhood. [Ahmad and Shin \(2021\)](#) proposes combining two chaotic maps with the Huffman coding technique to achieve data compression and encryption at the same time. The data is initially rearranged using a chaotic map, and the rearranged data's neighboring values are transformed uniquely using the Chinese Remainder Theorem (CRT). During the substitution phase, these altered values are compressed using the Huffman compressor, resulting in 8-bit outputs. This technique retains its lossless nature, rendering it particularly suitable for applications in medical image compression and encryption. [Li, Yu, Wang \(2021\)](#) came up with a new way to compress, authenticate, and secure images in the cloud using Compressive Sensing (CS). They suggested a new chaotic map called the Logistic-Tent-Sine chaotic map (LTSS), which has a wider chaotic range and better chaotic behavior than traditional 1D chaotic maps. In their encryption method, the original picture is split into two parts: 'low-frequency and high-frequency'. The low-frequency component is encrypted using a binary data cyclic encryption algorithm devised by the authors, while the high-frequency component is first randomized via CS-based compression and then encrypted using DRPE. Experimental evaluations have shown that the proposed scheme can withstand exhaustive attacks, differential attacks, and conventional attacks.

In 2022, [Maazouz et al. \(2022\)](#) propose a novel algorithm for image encryption security, introducing a chaos based system and utilizing its state-variables to generate a unique substitution matrix. This matrix plays a critical role in generating encryption keys and operating the encryption algorithm. Drawing upon components from 'the Advanced Encryption Standard' (AES) and sections of the Feistel network, the method incorporates several steps. Comprehensive testing showcases the algorithm's robustness against statistical and differential attacks, validating its effectiveness in image encryption. [Zhang and Zhang \(2022\)](#) introduce a multi-image encryption technique that combines chaos theory and gene fusion. The algorithm consists of multiple phases. Using a chaotic sequence, the pixel locations of k plain pictures are shuffled during initialization. Then, a chaotic picture that is generated from a chaotic system coordinates the periodic shifting of pixels across many images to produce their jumbled representations. The chaotic system generates a DNA rule controller that determines which encoding, operation, and decoding rules are applied to each picture. Finally, the gene fusion technique is used to accomplish diffusion between the multiple images, resulting in the generation of encrypted images. [Mondal and Singh \(2022\)](#) introduces a lightweight

image encryption method employing chaotic maps and diffusion circuits. A fundamental aspect of chaotic maps involves the creation of random number sequences that are used to permute and substitute pixel values in pictures. Interestingly, both substitution and permutation are executed simultaneously in a single picture scan, which reduces the complexity of the computation. Furthermore, the computational load is kept to a minimum by using simple bit-wise substitution procedures. The proposed system undergoes comprehensive statistical and security analyses to verify its resilience against attacks.

In 2023, [Alexan et al. \(2023\)](#) present a captivating algorithm for image encryption that combines the power of the KAA map with multiple chaotic maps. The security of the images is ensured using a novel use of bit confusion and diffusion, both of which are nods to Shannon's concepts. Confusion is heightened by employing two encryption keys: one derived from the 2D Logistic Sine chaotic map combined with a Linear Congruential Generator, and the other from the Tent chaotic map in conjunction with the Bernoulli map. Diffusion is masterfully executed through the application of the KAA map. Extensive evaluations have shown that the technique is effective in warding off a wide variety of cryptanalytic attacks, including brute-force, statistical, differential, and visual attacks. [Rani et al. \(2023\)](#) introduced an encryption method for grayscale and colored images that integrates a unique magic square and differential encoding technique with the 2D Arnold scrambling algorithm and PWLCM chaotic map. Differential encoding improves the security of individual bits by scrambling pixel positions, while the magic square modifies pixel values to disguise the original image information. Performing XOR operations on the encoded values and the values from the chaotic map, and then applying a circular shift, enhances the level of complexity. Rigorous analyses confirm the model's efficacy in defending against entropy, statistical, cropping, and differential attacks. [Alawida \(2024\)](#) introduced an innovative image encryption algorithm designed specifically for Industrial Internet of Things (IIoT) scenarios that have limited resources. The algorithm utilizes a 1-D perturbed logistic chaotic map. The algorithm employs both diffusion and permutation in a single round, hence decreasing processing time. It generates a data sequence that makes three matrices that are the same size as the image. Two of the matrices are used to change the order of the pixels, and the third is used for diffusion. These preprocessed matrices have the capability to encrypt numerous images using the same session key, hence improving efficiency. The experimental findings demonstrate that the model is highly resistant to cyberattacks such as histogram and differential analysis. Additionally, it efficiently encrypts many images with low latency in both wired and wireless communications, making it suitable for practical implementation in IIoT applications. [Shao et al. \(2023\)](#) introduced a new PLSM chaotic map that combines logistic and sine maps to enhance chaotic behavior and ergodicity while maintaining low temporal complexity. In addition, they created an image encryption system using PLSM and random exclusive OR diffusion. The initial values and parameters of PLSM are determined using a key distribution approach, which involves a 256-bit secret key. PLSM generates four chaotic sequences, which are then used in three rounds of random exclusive OR diffusion to protect the image information. The simulation results on several images have shown that the technique performs well and has a decreased time complexity. Furthermore, security assessments have demonstrated that the suggested system offers a superior level of security in comparison to various picture encryption algorithms that are currently in use. A concise comparative analysis of the discussed literature is provided in the following [Table 2](#).

3. Prefatory material

3.1. Skew Tent Map

A number of one-dimensional chaotic maps are useful for producing digital noise in security applications ([López-Hernández et al., 2012](#);

Table 2

Comparative analysis of literature survey.

Paper title	Used chaotic map	Characteristic of chaotic map	Overview	Positive aspects	Limitations
Color Image Encryption Algorithm Based on Dynamic Chaos and Matrix Convolution (Hu et al., 2020)	Fibonacci chaotic system	–	The paper proposes a color image encryption algorithm using a cloud model Fibonacci chaotic system combined with matrix convolution enhancing security and randomness for robust encryption.	High security, enhanced randomness, efficient performance, robust key sensitivity, and low pixel correlation.	Complex implementation, high computational overhead, no parallel processing, and dependency on initial parameters.
Medical image encryption using fractional discrete cosine transform with chaotic function (Kumar et al., 2019)	Logistic map & Chebyshev map	–	The research presents a medical image encryption algorithm that utilizes the fractional discrete cosine transform (FrDCT) in conjunction with a chaotic map. This algorithm aims to improve the security of medical data during transmission and storage.	Minimal pixel correlation, strong key sensitivity, big keyspace, effective performance, and good security.	Requires significant processing resources and lacks adaptability for handling large-scale datasets.
A secure image encryption scheme based on cellular automata and chaotic skew tent map (Mondal et al., 2019)	Skew Tent Map	Short chaotic range and low LE value	The research presents a secure picture encryption system that utilizes a chaotic skew tent map and cellular automata (CA) to achieve efficient and strong encryption.	Robust security measures, rapid creation of pseudo-random number sequences (PRNS), extensive range of possible encryption keys, efficient operational performance, and minimal correlation between pixels.	Potential vulnerability to noisy channels, high computational burden, and complex implementation.
The fast image encryption algorithm based on lifting scheme and chaos (Zhang, 2020)	PWLCM	–	The research provides a cloud-based secure digital locker using a Paillier-based difference function and chaos-based cryptosystem for image authentication and encryption that preserves anonymity.	Protects user identity and data privacy, improves security with bit-level picture confusion and chaotic maps, and supports encrypted authentication.	Complex implementation, high computational burden for cloud servers, potential vulnerability to internal attacks, issues with energy efficiency, image sharing security, key management, and dependency on cloud infrastructure.
Privacy preserving cloud based secure digital locker using Paillier based difference function and chaos based cryptosystem (Falmari & Brindha, 2020)	Arnold's cat map & Chebyshev map	–	The study introduces a safe digital locker system that preserves confidentiality in the cloud. It utilizes a difference function based on the Paillier encryption scheme and a cryptosystem based on chaos theory to ensure secure image authentication and encryption.	Protects user identity and data privacy, improves security with bit-level picture confusion and chaotic maps, and supports encrypted authentication.	Complex implementation, high computational burden for cloud servers, potential vulnerability to internal attacks, issues with energy efficiency, image sharing security, key management, and dependency on cloud infrastructure.
An Efficient Image Encryption using Deep Neural Network and Chaotic Map (Maniyath & V, 2020)	–	–	Deep neural networks are used to optimize simple encryption algorithms with a chaotic map for improved security performance.	High security performance with no negative effect on image quality, robust key generation using stacked auto-encoders, and efficient resistance to various attacks through optimized deep learning and chaotic maps integration.	The implementation is computationally expensive and sophisticated, consumes a lot of memory, and does not cover all possible attack situations.

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Table 2 (continued).

IESCA: An efficient image encryption scheme using 2-D cellular automata (Roy et al., 2021)	2-D Moore cellular automata as chaotic map	–	The study presents IESCA, a lightweight image encryption system that utilizes 2-D Moore Cellular Automata (MCA) to ensure safe picture transmission in IoT contexts.	Strong security measures, optimal efficiency, minimal resource usage, extensive randomness, and encryption without any loss of data.	The implementation's complexity, reliance on starting parameters, possible computing cost, and scalability problems with big images or datasets are its limitations.
A novel hybrid image encryption–compression scheme by combining chaos theory and number theory (Ahmad & Shin, 2021)	2D hyper-chaos discrete nonlinear dynamic system and Logistic Map	Broader Chaotic range but low LE value	The paper presents a method for encrypting and compressing medical images using a combination of chaos theory and number theory. The method involves using chaotic permutation and substitution techniques followed by Huffman coding to ensure both security and efficiency in the processing of medical images.	Enhanced security, effective and error-free compression, superior performance, and adaptability for secure transfer and storage.	May require a substantial amount of computer resources, relies on the initial settings, and may be susceptible to interference from noisy channels.
A Novel 1D Chaotic System for Image Encryption Authentication and Compression in Cloud (Li, Yu, Wang, 2021)	Logistic-Tent-Sine System	Larger chaotic ranges, better chaotic behaviors but Low LE value	The study presents a new approach for image compression, authentication, and encryption in cloud environments. This algorithm utilizes a unique 1D chaotic system called the Logistic-Tent-Sine System. It combines compressive sensing with chaotic maps and palmprint features to achieve authentication.	Heightened security, effective data compression, strong authentication, enhanced reconstruction quality, and decreased computational load on user devices.	The restrictions include energy efficiency, image sharing security, key management, and the necessity for near-real application comparisons with other algorithms. It is also complicated, computationally intensive for cloud servers, vulnerable to internal threats, and dependent on cloud infrastructure.
FPGA implementation of a chaos-based image encryption algorithm (Maazouz et al., 2022)	Linear discrete time systems	–	The research presents a novel image encryption technique based on chaos, which is implemented on FPGA. The algorithm incorporates a new chaotic system and combines the Feistel network with stages from AES to provide improved security in IoT and other embedded networks.	Low correlation with high entropy, improved cryptographic characteristics, excellent security, and efficient performance.	Intricate execution, substantial computational cost, reliance on starting parameters, and resource-intensive.
Multiple-image encryption algorithm based on chaos and gene fusion (Zhang & Zhang, 2022)	Logistic Map & Rossler Chaotic system	Broader chaotic range with a high sensitivity to its initial values but low LE values	To improve security and efficiency for huge data transmission, this research suggests a multiple-image encryption technique that uses gene fusion and chaos.	Robust security measures, effective encryption techniques, high-quality encryption, minimal pixel correlation, and capacity to handle many photos with ease.	Limited flexibility for very big datasets, substantial processing overhead, complex implementation, and reliance on starting settings.
A lightweight image encryption scheme based on chaos and diffusion circuit (Mondal & Singh, 2022)	Chaotic Standard Map (CSM)	–	Using diffusion circuits and chaotic maps, the research suggests a low-power image encryption method for Internet of Things devices that ensures effective and safe image encryption.	Robust security measures, optimal operational efficiency, streamlined implementation, heightened key responsiveness, and consistent histogram patterns.	There is a possibility of experiencing computational burden when dealing with huge photos, and there may be challenges in terms of scalability.

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Table 2 (continued).

Color Image Encryption Through Chaos and KAA Map (Alexan et al., 2023)	2D Logistic Sine Map, Tent Map, Bernoulli Map	Strong chaotic behavior, excellent randomness, and ergodicity but small chaotic range	The study offers a color picture encryption technique that uses the KAA map and several chaotic maps to improve security while adhering to Shannon's confusion and diffusion rules. It is very resistant to various cryptographic attacks and passes all NIST SP 800 tests.	High security and resilience, enhanced confusion and diffusion via 1D and MD chaotic functions, a huge key space of 2478 for high resistance to brute-force assaults, and efficient performance with superior metrics when compared to cutting-edge approaches.	Multiple chaotic maps and key generation result in increased complexity and computational cost, as well as implementation issues that necessitate careful handling of numerous factors and beginning circumstances.
Image encryption model based on novel magic square with differential encoding and chaotic map (Rani et al., 2023)	2D Arnold map & PWLCM	–	The research offers an image encryption model employing a unique magic square, differential encoding, and a chaotic map with 2D Arnold scrambling for confusion and piece-wise linear chaotic map for diffusion.	High security, solid performance, novel methods, effective confusion and diffusion, and grayscale/color image adaptability.	Complex implementation, substantial computational overhead, and data recovery concerns after major information loss.
A Novel Image Encryption Algorithm Based on Cyclic Chaotic Map in Industrial IoT Environments (Alawida, 2024)	Cyclic Chaotic Map	Large chaotic area, high LE value	The paper presents an image encryption algorithm for Industrial IoT (IIoT) environments using a cyclic perturbed logistic map (CPLM) to enhance security and efficiency through single-round encryption.	Enhanced security, high key sensitivity, efficient single-round encryption, versatility for different image sizes, and low computational burden for IoT sensors.	Complex implementation, high computational overhead for cloud servers, potential internal attack vulnerability, reliance on cloud infrastructure.
Chaotic Image Encryption Using Piecewise-Logistic-Sine Map (Shao et al., 2023)	Piecewise-Logistic-Sine Map	Better chaotic behavior, low time complexity	The research presents a novel image encryption strategy that employs a newly developed Piecewise-Logistic-Sine Map (PLSM) to achieve improved chaotic behavior and reduced time complexity. This scheme utilizes a 256-bit secret key and incorporates three rounds of random XOR diffusion.	Improved security, more unpredictable behavior, efficient time complexity, effective performance, and heightened key sensitivity.	The implementation is complex, requires significant computational resources, lacks a scrambling step, and relies on the initial parameters.

Machicao & Bruno, 2017; Xu et al., 2010). These chaotic maps are desirable because they can produce number sequences with statistical distributions that closely approximate the uniform function. Skew Tent chaotic Map (STM) is an example of these maps; This is a simple 1-D function ($f(x): [0,1] \rightarrow [0,1]$) (Palacios-Luengas et al., 2019), defined as:

$$y_{n+1} = \begin{cases} \frac{y_n}{p}, & 0 < y_n \leq p \\ \frac{1-y_n}{1-p}, & p < y_n < 1 \end{cases} \quad (1)$$

where $p \in (0, 1)$. As per Eq. (1), the described transformation maintains continuity and displays piecewise linearity, showcasing distinct linear regions within the intervals $[0, p]$ and $[p, 1]$ (Hasler & Maistrenko, 1997). It is vital to note that the beginning value ' y_0 ' as well as the parameter ' p ' have a significant impact on the subsequent sequence. Even minor deviations in these values can yield entirely different sequences, highlighting the sensitivity to its seed and parameter values—a hallmark trait of the chaotic systems. As indicated by the bifurcation diagram, it can be observed that the STM exhibits chaotic behavior within the range $(0, 1)$. Moreover, the consistently positive Lyapunov Exponent (LE) of the STM signifies heightened sensitivity to initial values and parameters, and showcases intricate chaotic dynamics. Fig. 3 provides a visual depiction of the STM's sensitivity analysis, Lyapunov exponent (LE), and bifurcation diagram.

3.1.1. Small chaotic region

The chaotic behavior of the skew tent map is observable only within the parameter range of $[0, 1]$. Outside of this interval, the map's dynamics become less unpredictable and less chaotic. Even though this area is small, it is an important place to study and learn about chaotic systems.

3.1.2. Sensitivity to its initial seed values

The skew tent map exhibits a positive but relatively small Lyapunov Exponent (LE) within the interval $[0, 1]$. Although it is sensitive to initial conditions, the divergence between two sequences with slightly distinct initial values is not discernible until roughly forty iterations. This suggests that the skew tent map is less sensitive to its initial conditions, resulting in a delayed rate of divergence with time.

3.1.3. Chaos annulling

Many chaotic maps have regions inside their domains where chaotic behavior is suppressed by special conditions called chaos-annulling traps. In the case of the Skew Tent Map (STM), it has been identified that there are specific trap conditions where irregular behavior is not present. As described in Palacios-Luengas et al. (2018), these trap conditions for the STM include particular values of the initial condition x_0 such as (i) 0, (ii) 1, (iii) $\frac{1}{2-p}$, (iv) p , (v) $\frac{p}{2-p}$. In addition, when p is set to 0.5, the trap condition applies to any value of x_0 . In Fig. 4, we can see the path of STM when $x_0 = 0.2$ and $p = 0.5$. After a few cycles, the state will be stable.

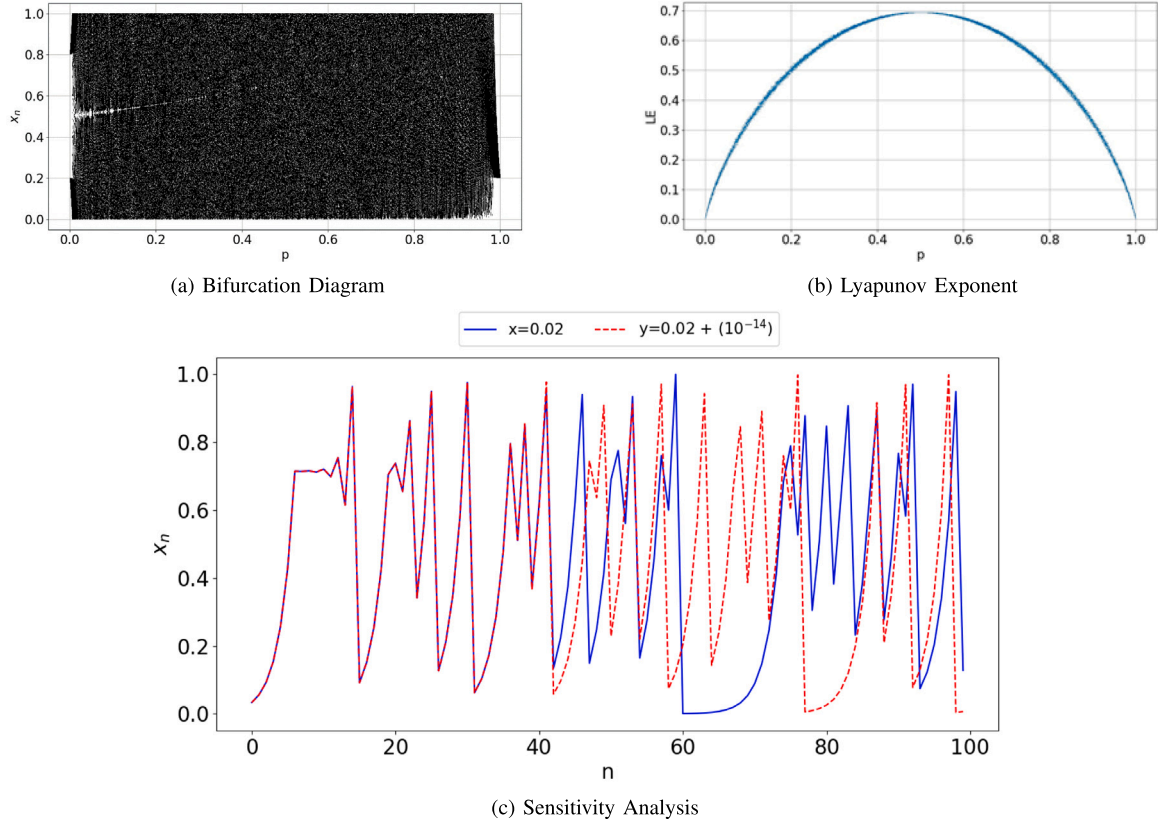
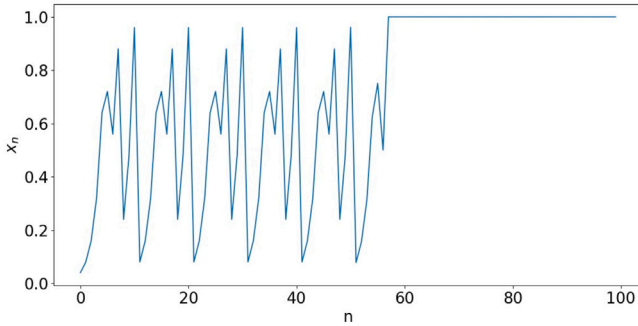


Fig. 3. Analysis for Skew Tent Map (STM).

Fig. 4. The trajectory of the Skew Tent Map when $p = 0.5$.

4. The proposed Sine Skew Tent Map (S-STM)

The sine function, being a “periodic trigonometric function with limited orbits and intricate nonlinear traits”. When added to any 1-D chaotic map, it could make the unpredictable behavior even worse. The chaotic map’s unpredictability can be further enhanced through the utilization of techniques termed “perturbations”. Considering these aspects, a new version known as the Sine Skew Tent Map (S-STM) is introduced. The mathematical expression defining the S-STM is given by Eq. (2):

$$x_{n+1} = \begin{cases} \sin\left(\left(\frac{1-p}{1+p} - \frac{\pi x}{1-p}\right) \cdot 10^3\right), & -1 \leq x_n \leq p \\ \sin\left(\left(\frac{1+p}{1-p} - \frac{\pi(1-x)}{1-p}\right) \cdot 10^3\right), & p < x_n \leq 1 \end{cases} \quad (2)$$

In the proposed piecewise equation, the variable x_0 represents the initial state, while p serves as the control parameter. Both x and p are restricted to the range of $[-1, +1]$ excluding the value ‘0’. To

enhance the complexity and unpredictability of the sequence, a perturbation technique involving multiplication by 10^3 is employed. In the subsequent section, a domain extension for the intended S-STM was conducted, Confirming the chaotic attributes of the recently acquired map.

4.1. Performance assessment

In this section, we examine the Sine Skew Tent Map’s (S-STM) chaotic properties and performance. We use popular evaluation techniques, such as the sensitivity to initial seed values, the Lyapunov Exponent, and bifurcation diagram, to examine the chaotic behavior of the suggested system. To demonstrate the improved chaotic qualities of the proposed S-STM, we next compare its evaluated findings with those of the STM and other current chaotic maps.

4.1.1. Bifurcation diagram

The chaotic behavior of chaotic maps is determined by their control parameters. However, the range of parameters that result in genuine chaos can be constrained, resulting in a restricted key space. This restriction is detrimental to cryptographic applications. The bifurcation diagram for the proposed Sine Skew Tent Map (S-STM) is shown in Fig. 5(a), depicting the range of control parameters where chaotic behavior occurs. The S-STM exhibits unpredictable behavior in the interval $[-1, +1]$, excluding 0. For cryptography applications, a wider range of chaotic parameters is preferred since it improves resilience to exhaustive search attacks. This expanded spectrum ensures that cryptographic applications are more secure and efficient.

4.1.2. Lyapunov Exponent (LE)

The utilization of chaotic maps in pseudorandom number generators (PRNGs) and cryptographic applications depends on how sensitive they are to their initial seed values. This sensitivity is measured in a

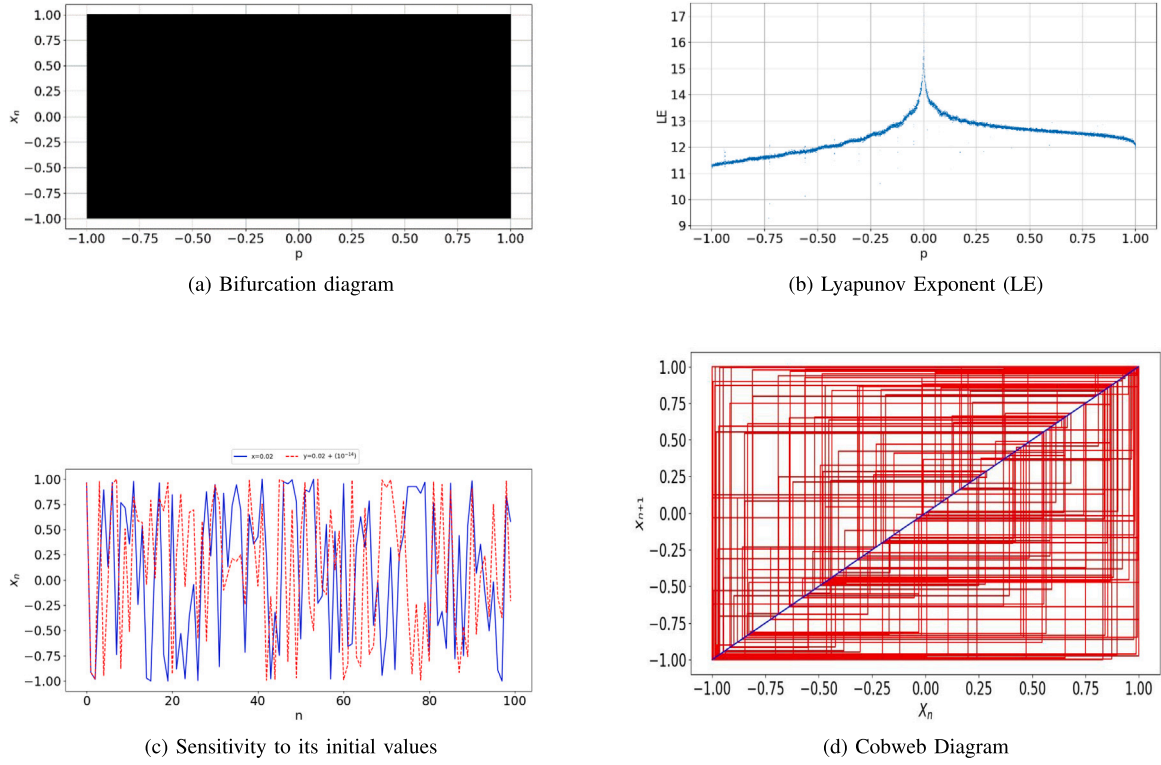


Fig. 5. Analysis for Sine Skew Tent Map (S-STM).

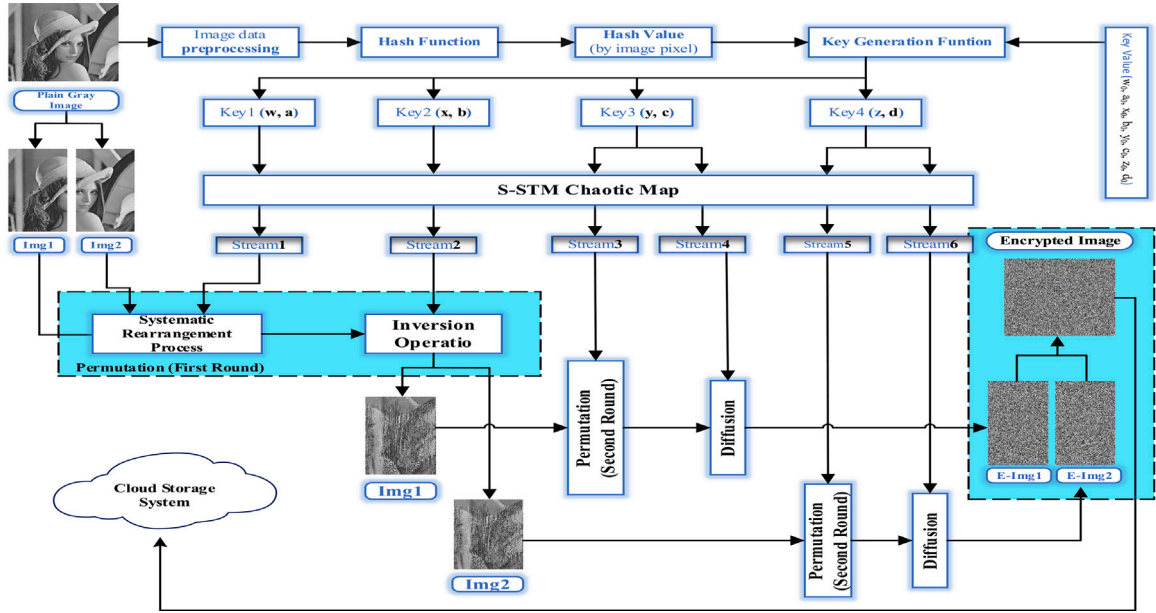


Fig. 6. The schematic representation of the proposed image encryption scheme for cloud storage.

chaotic map using the Lyapunov exponent (LE). A positive Lyapunov exponent (LE) implies that after a certain number of iterations, the map's trajectory will diverge. Furthermore, a higher LE denotes a more complex system with unpredictable behavior.

$$LE = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \ln |f'(x_i)| \quad (3)$$

To calculate the Lyapunov Exponents for the suggested map, we utilize Eq. (3) specified in Wolf et al. (1985). The corresponding LE

values are showcased in Fig. 5(b). These outcomes consistently demonstrate that the Lyapunov exponent of proposed chaotic system remains positive and notably high across all parameter values for the interval $(-1, +1)$.

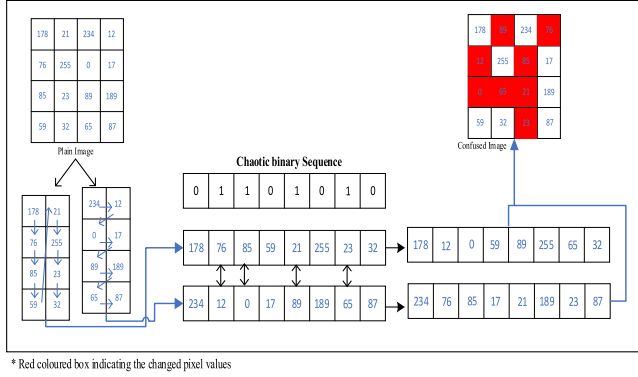
4.1.3. Sensitivity to initial conditions

Chaos theory is renowned for its sensitivity to initial seed values, signifying that even a slight alteration in these values can lead to a significant change in the overall results. To verify this property, the S-STM chaotic map was tested with two distinct initial seed values, x_0

Table 3

Comparative analysis of S-STM against several existing chaotic maps based on their Maximum Lyapunov Exponent (MLE) and chaotic region.

Reference	Chaotic map	Chaotic region	Maximum LE
Murillo-Escobar et al. (2017) and Silva and Peacock-Lopez (2017)	Logistic map	[3.569, 4]	≈ 0.6724
Elmanfaloty and Abou-Bakr (2019)	Skew Tent Map	[0, +1]	≈ 0.7
Zhao et al. (2019)	Lorenz system	–	≈ 2.2991
Zhang et al. (2020)	Piecewise Cubic Map	[0, 4]	≈ 5.4678
Cang et al. (2021)	Generalized Sprott-A system	[0, 2]	≈ 0.9
Akraam et al. (2023)	Modified Tent Map	[0, 2]	≈ 1.23
Mfungo et al. (2023)	Sigmoid Logistic Map	[5, 6]	≈ 3.63
Proposed work	Sine Skew Tent Map	$[-1, +1) - 0$	≈ 15.98

**Fig. 7.** Systematic rearrangement process.

$= 0.02$ and $x_0 = 0.02 + 10^{-14}$, while the parameter ‘p’ value remained constant at 0.6. As depicted in Fig. 5(d), the resulting trajectories result in orbits that are dramatically dissimilar. This demonstrates that even minor modifications to the initial conditions of the S-STM chaotic map can generate distinct sequences, providing valuable insight into the map’s chaotic behavior.

4.1.4. Cobweb diagram

A cobweb plot is a way to look at how functions change over time in dynamical systems (Tomida, 2008). It can also be used to track the long-term behavior of a dynamic system and see how a function changes over and over again in a single graph. When the repeated track of a chaotic system does not match up in this parameter space, the system is more dynamic, and the order it makes is random. Fig shows cobweb plot graphs for $x_0 = 0.20$ and $r = 0.34$. The analysis of the cobweb diagram revealed that the trajectories of the series generated by the S-STM are not coincident, indicating that the S-STM has a high level of performance.

As the previous findings reveal, the suggested SSTM chaotic map displays a bigger chaotic space. Furthermore, with a larger LE value and better sensitivity to beginning values, it looks to outperform the standard STM. Table 3 (Akraam et al., 2023; Cang et al., 2021; Elmanfaloty & Abou-Bakr, 2019; Mfungo et al., 2023; Murillo-Escobar et al., 2017; Silva & Peacock-Lopez, 2017; Zhang et al., 2020; Zhao et al., 2019) compares the S-STM map’s Lyapunov exponent and chaotic region to those of other existing chaotic maps.

5. Proposed image encryption approach

In this section, we will use the proposed Sine Skew Tent Map (S-STM) technique to develop a novel image encryption scheme called SSTM-IES (S-STM Based Image Encryption Scheme) for cloud storage system.

5.1. Overall structure

The architecture of SSTM-IES is depicted in Fig. 6. Initially, a secret key generation process is employed to produce the initial seed values, termed keys, for the S-STM. Subsequently, S-STM generates six chaotic matrices using these four keys. Then, the image encryption procedure begins with image splitting, in which the pixels of the provided picture I are partitioned into two different segments, I_1 and I_2 , during a predefined period.

In the first permutation round, which occurs following the image-splitting step, the method utilizes Systematic rearrangement process and Inversion operations to facilitate pixel exchange between I_1 and I_2 , as well as to disperse pixels across the image. This process is instrumental in fortifying the encryption scheme’s robustness. Additionally, the two image halves will undergo a secondary round of permutation, succeeded by a diffusion process. This permutation and diffusion process is executed with the aid of four distinct chaotic sequences. The incorporation of these additional permutation and diffusion steps significantly enhances the overall effectiveness of the encryption scheme. The complete encryption process is encapsulated in Algorithm 1, providing a comprehensive description of all the steps involved.

5.2. Initial key generation

In Alvarez and Li (2006), suggests that a chaos-based encryption algorithm must use a key length exceeding 100 bits to counteract potential brute-force attacks. To meet the stringent security requirements and align with the inherent structure of SSTM-IES, the length of the secret key ‘ $k = (w_0, x_0, y_0, z_0, a_0, b_0, c_0, d_0)$ ’ has been established at 424 bits. In the key k, (w_0, x_0, y_0, z_0) are the initial values with size of 53 bits, and a_0, b_0, c_0, d_0 are the control parameters of chaotic map S-STM with size of 53 bits.

$$\begin{cases}
 p = k_1 + k_2 + k_3 + \dots + k_{32} \\
 val = \frac{p}{32 \times 256} \\
 m = w_0 + x_0 + y_0 + z_0 + a_0 + b_0 + c_0 + d_0 \\
 w = (m + w_0 + \frac{val}{3}) \% 1 \\
 x = (x_0 + \frac{w}{3}) \% 1 \\
 y = (y_0 + \frac{x}{3}) \% 1 \\
 z = (z_0 + \frac{y}{3}) \% 1 \\
 a = (a_0 + \frac{val}{3}) \% 1 \\
 b = (b_0 + \frac{a}{3}) \% 1 \\
 c = (c_0 + \frac{b}{3}) \% 1 \\
 d = (d_0 + \frac{c}{3}) \% 1
 \end{cases} \quad (4)$$

Using a separate system key can potentially make the system more susceptible to chosen plaintext attacks. To mitigate this risk, the SHA3-256 function is applied to the pixels of the original image to derive a hash value, which is then employed to update the initial key. This process effectively lowers the system’s vulnerability to chosen plaintext attacks.

Algorithm 1**Step 1:**

Convert I matrix into 1D matrix M by concatenating the image pixel values

Step2: Key Generation

Initialize an empty string P.

for each element M[i] in the key **do**

Append M[i] to P.

Calculate P as the SHA3-256 hash of P.

end for

Execute Eq. (4) to generate modified keys: $k_1 = (w, a)$,

$k_2 = (x, b)$, $k_3 = (y, c)$, and $k_4 = (z, d)$.

Step3: Image Preprocessing

if I is a colored image **then**

Convert it into grayscale image IG of dimension $(H \times 3W \times D)$.

else

Set IG to be equal to I.

end if

Convert IG into a 1D vector of pixels and split it into two vectors I_1 and I_2 of size $(H \times W)/2$ each.

Step 4: Chaotic Sequence Generation

Execute Algorithms 2, 3, 4 to generate chaotic sequences.

Step 5: Permutation (First Round)

Convert I_1 and I_2 matrices into 1D matrices by concatenating the image pixel values in a column and row-wise manner, respectively.

for each element in the chaotic sequence S_4 **do**

if $S_4[i] = 1$ **then**

Interchange the I_{th} pixel values of I_1 and I_2 .

end if

end for

Convert the matrices back into 2D matrices.

for each column i in the range of $(2 \times W)$ **do**

Get p from $S_3[i]$.

Get q from $S_3[i+1]$.

Reverse the pixels within the range $[p, q]$ in ith column.

end for

Step 6: Permutation (Second Round)

Convert I_1 and I_2 matrices into 1D matrices.

Sort the chaotic sequences and save their indexes.

Rearrange the pixels of both matrices according to their sorted chaotic sequences.

Step 7: Diffusion

Execute Algorithm 5 on both matrices for diffusion.

Step 8: Output

The encrypted image I_E is obtained as the result.

Algorithm 2

$L_1 = (H \times W \times D) / 2$

Execute Eq. (2) to generate the Chaotic Sequence S_4

for i in range of (L_1)

$S_4[i] = [\text{floor}(S_4[i] \times 10^3)] \% 2$

Output: S_4

The resultant hash value is partitioned into 32 blocks, with each block being converted into a decimal number, represented as $k_1, k_2,$

$\dots, k_{32}$. Then, the initial key k is set to ' $k_1 = (w_0, x_0, y_0, z_0)$ ' and ' $k_2 = (a_0, b_0, c_0, d_0)$ '. And, Using the key generation Eq. (4), the initial key will be modified.

The updated key will yield four distinct key values, $k_1 = (w, a)$, $k_2 = (x, b)$, $k_3 = (y, c)$, and $k_4 = (z, d)$.

5.3. Image pre-processing

Initially, the system receives an input plain picture with dimensions $H \times W \times D$, where H represents the image's height, W denotes its width, and D specifies the image depth. For grayscale images, D is assigned a value of 1, while for color images, it is set to 3. In the case of a color image ($D = 3$), it is first converted into a grayscale image of size $H \times 3W \times D$, with D set to 1. Subsequently, the grayscale image undergoes a transformation to be represented as a 1D matrix. This transformation is achieved by concatenating the pixel values in a column-wise manner, resulting in a linear format representation of the image. This matrix is subsequently divided into two segments. These two 1D matrices are then restructured into 2D matrices, forming segments I_1 and I_2 of the original image I. Now the both matrices of the image I_1 and I_2 will be go through first round of permutation.

Algorithm 3

$L_1 = (H \times W \times D) / 2$

Execute Eq. (2) to generate the Chaotic Sequence S_4

for i in range of (L_1)

$S_4[i] = [\text{floor}(S_4[i] \times 10^3)] \% 2$

Output: S_4

5.4. First permutation round

In the proposed encryption method, permutation plays a fundamental role by reordering pixels within an image while preserving their initial values. To significantly enhance the perturbation within Image I, we incorporate two rounds of permutation. In the initial process of first permutation round, a methodology inspired by sophisticated techniques for achieving specific objectives is utilized. By utilizing the derived key $k_4 = (z, d)$, a chaotic sequence, referred to as S_4 , is generated. S_4 has a length of $L_1 = (H \times W \times D)/2$ and is subsequently converted into a binary representation through the application of Algorithm 2, serving as the basis for the permutation process. Employing the binary values from S_4 , the pixels within matrices I_1 and I_2 undergo a systematic rearrangement process, as visually represented in Fig. 7. This procedure effectively redistributes the pixels throughout the entire image.

To further enhance the complexity of the image, an additional operation is introduced. This operation generates another chaotic sequence, denoted as S_3 (with a length of $W \times 2$), by utilizing the previously computed key $k_3 = (y, c)$. Subsequently, S_3 is transformed into an integer sequence through the application of Algorithm 3. Following this transformation, an 'Inversion Operation' is applied. This operation selects consecutive numbers from S_3 to bring about changes in a single column of the matrix, as exemplified in Fig. 8.

5.5. Second permutation round

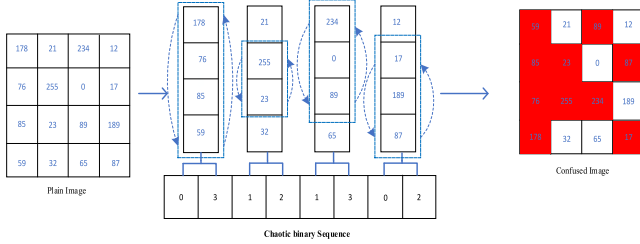
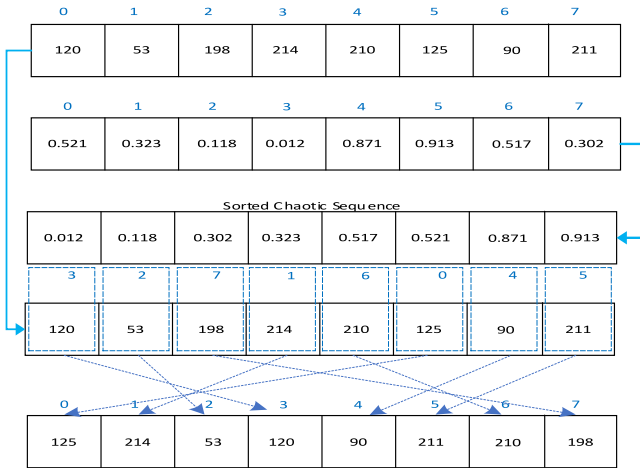
During the second permutation round, we employ a conventional permutation technique based on the chaotic map for both segments of image I. Our approach encompasses transforming the half-image matrices with dimensions $[H \times W/2]$ into 1D matrices of dimension $[1 \times (H \times W/2)]$. The image matrices I_1 and I_2 undergo scrambling through the utilization of two distinct chaotic sequences, C_1 and C_2 , each having a length of $L_p = H \times W/2$. These chaotic sequences

Algorithm 4

```

Execute Eq. (2) to generate the Chaotic Sequence 'C'
C ← Chaotic Sequence
Ld ← Length of Chaotic Sequence
Cd ← [] × Ld #(Array of length Ld)
for i in range of (Ld)
    p=C[i];
    p = [p × (105)] % 256
    Cd[i] = p;
Output: Cd

```

**Fig. 8.** Inversion operation.**Fig. 9.** Permutation process.

are generated via two different keys obtained in the previous section, namely $k_1 = (w, a)$, $k_2 = (x, b)$, respectively. Subsequently, the chaotic sequences are orderly arranged in an ascending manner, and their corresponding indices are recorded. As illustrated in Fig. 9, the elements of the 1D array are then reorganized based on the sorted order of the chaotic sequence.

5.6. Diffusion

Diffusion is a crucial step where the original pixel values of the image undergo modification. To enhance the security and complexity of the encryption method, an additional diffusion process is applied to the previously permuted pixels. During this diffusion process, the chaotic sequences C_1 and C_2 , generated in the permutation phase, are transformed into new chaotic sequences, Cd_1 and Cd_2 , using Algorithm 4. These transformed chaotic sequences produce two sets of integer chaotic sequences, which are then utilized to execute the diffusion process on image matrices I_1 and I_2 , respectively. The diffusion process follows the guidelines specified in algorithm 5. After the permutation

Algorithm 5

```

I ← Image 1D matrix #(Array of length Ld)
Execute Eq. (2) to generate the Chaotic Sequence 'C'
C ← Chaotic Sequence
Id ← [] × (length(I))
for i in range of (length(I))
    if i=0;
        t=I[last];
    else:
        t = I[i-1]; Id[i] ← I[i] XOR C[i] XOR t
Output: Cd

```

and diffusion processes have been applied to both parts of the image, the 1D arrays are reverted to their original dimensions as image matrices. These permuted and diffused matrices are then merged together to create the encrypted image.

6. The decryption algorithm

During the decryption process, we start with the encrypted image and its corresponding hash value, denoted as $h_1, h_2, h_3, \dots, h_{64}$. Using the initial key values and this hash value, we generate four secret keys using Eq. (4). We then generate chaotic sequences in the same way as we did during the encryption process. Next, we divide the encrypted image into two parts and apply reverse diffusion and undo the second round of permutation operations to each part. These steps effectively reverse the encryption process.

After that, we perform the first permutation operations, similar to what we did during the encryption process, which are crucial for recovering the original plain image. Finally, by merging the two parts of the image that were split earlier, the decryption process successfully restores the plain image, ensuring both data integrity and security.

7. Experimental result and analysis

To assess the performance of the proposed encryption method in a cloud environment, we conducted a simulation experiment using a cloud simulation tool called CloudSim (Jayapandian, 2021; Umar et al., 2023). This tool, built on Java, proved highly effective for emulating cloud encryption and scheduling algorithms. Our experimentation took place on an HP Pavilion 15-cc102tx Laptop, which boasts 8 GB of RAM, 1 TB of storage capacity, a 1.60-GHz Intel Core i5 processor, and is equipped with the Windows 11 Pro operating system.

To validate the robustness of the developed approach, a range of attack analyses were conducted. These encompassed statistical analysis, differential analysis, assessment of encryption quality, evaluation of correlation and entropy, scrutiny of key space, assessment of key sensitivity, and chosen-plaintext attacks. The evaluation involves nine images: one color and eight grayscale images (including two medical samples - chest X-ray and brain MRI). The inclusion of the color and medical image aims to test the scheme's compatibility with these types of images. Fig. 10 illustrates the original images, their encrypted forms, and the corresponding decrypted forms. Table 4 provides a comprehensive summary of the statistical properties of the test images.

7.1. Analysis of histogram

The histogram of image gray values serves as an important indicator, reflecting the distribution of pixels within an image. Generally, plaintext images represent the observable real world and, as a result, exhibit a non-uniform histogram distribution for their grayscale images.

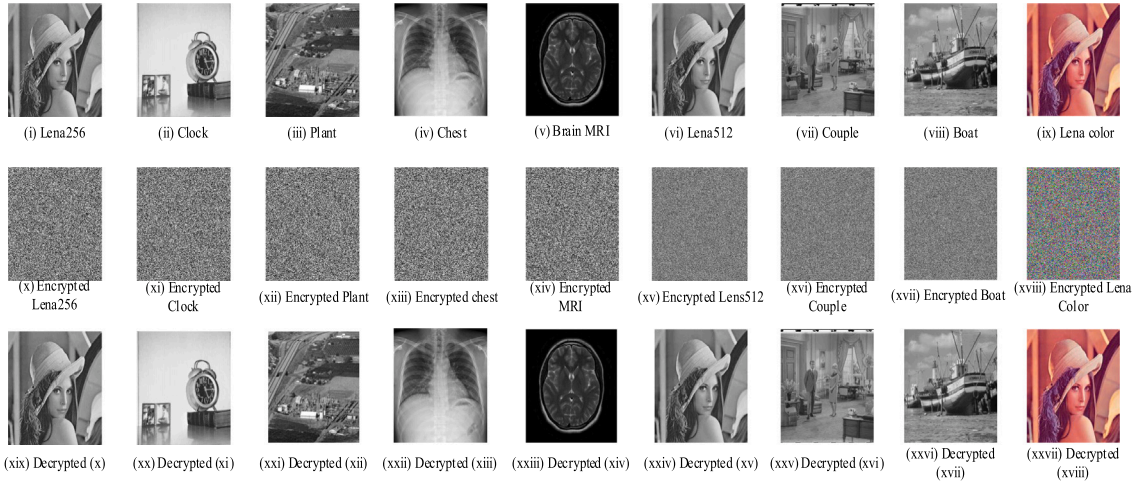


Fig. 10. Image encryption/decryption effects: ((i)–(ix)) Plain images, ((x)–(xviii)) Encrypted form of Corresponding images, and ((xix)–(xxvii)) Corresponding images after decryption.

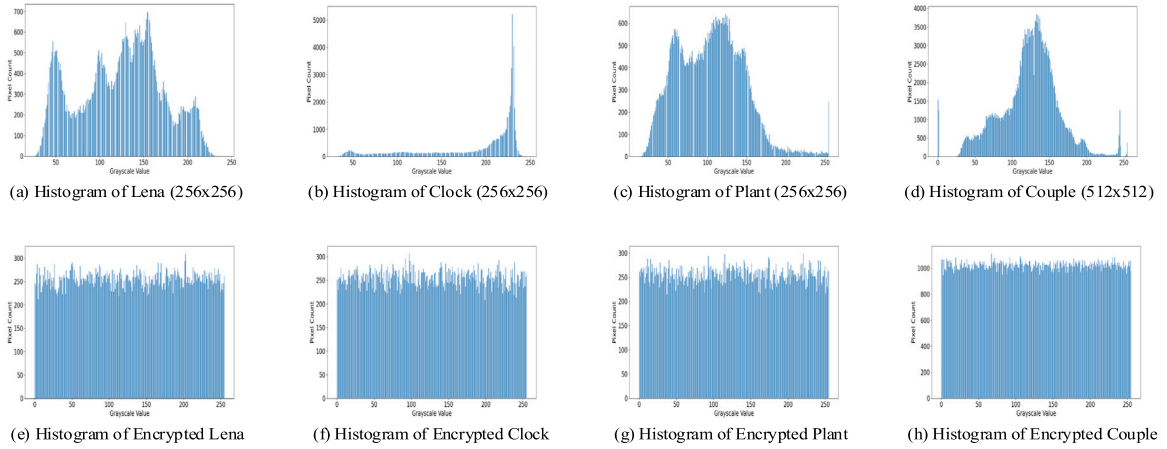


Fig. 11. Histogram analysis: (a–d) Plain images histograms, (e–h) Encrypted images histograms.

Table 4

Characteristic attributes of plaintext images from a statistical perspective.

S. No	Image	Resolution	HC	VC	DC	Entropy
(i)	Lena	256 × 256	0.95596	0.92123	0.89984	7.44286
(ii)	Plant	256 × 256	0.91223	0.94640	0.86163	7.34243
(iii)	Clock	256 × 256	0.97271	0.95465	0.93630	6.70566
(iv)	Lena	512 × 512	0.97797	0.96468	0.94673	7.44506
(v)	Couple	512 × 512	0.89072	0.88831	0.81041	7.20100
(vi)	Boat	512 × 512	0.94690	0.80621	0.80578	7.19137
(vii)	Chest	256 × 256	0.93437	0.92163	0.85456	5.40181
(viii)	Brain MRI scan	256 × 256	0.95624	0.95056	0.91750	5.41101
		R	0.97703	0.95545	0.93260	7.23528
(ix)	Lena	256 × 256 G	0.96732	0.94395	0.91888	7.56828
	(Color)	B	0.94547	0.92246	0.89417	6.91756

In contrast, for encrypted images, a chaotic visual effect is desired, necessitating a uniform histogram distribution to prevent attackers from extracting meaningful information. Fig. 11 displays the histograms of both plaintext and ciphertext of some images. The ciphertext images exhibit visually uniform histograms, clearly distinct from the histograms of the plaintext images. However, relying solely on visual inspection is inadequate for assessing the scheme's performance. A more practical metric to evaluate the scheme is the variance observed in the histograms (Zhang & Wang, 2014), expressed as follows:

$$Var(Z) = \frac{1}{x^2} \sum_{i=1}^N \sum_{j=1}^N \frac{1}{2} (Z_i - Z_j) \quad (5)$$

In this study, the vector Z represents the histogram values, defined as $Z = z_1, z_2, \dots, z_{256}$, where z_i and z_j represent the count of pixels with gray values i and j , respectively. The variance of histograms, $Var(z)$, serves as a crucial metric for assessing the uniformity of ciphered images. Decreased variance values indicate heightened uniformity in the distribution of pixel intensities. Table 5 presents the variances of histograms for both plaintext and ciphered images. The outcomes indicate considerably lower variances of histograms for the ciphered images in comparison to their respective plaintext counterparts. This suggests a heightened level of visual chaos and uniformity within the encrypted images.

Table 5
Variance of histograms of plaintext and encrypted images.

S. No	Image	Plain image	Cipher image
01	Lena 256	31 101.83	244.66
02	Plant	49 325.19	242.46
03	Clock	320 969.22	253.89
04	Lena 512	516 069.65	1017.56
05	Couple	1 200 497.33	1015.64
06	Boat	1 537 773.61	1012.87
07	Chest Xray	19 895.24	258.16
08	Brain MRI	1 868 091.74	245.85
	R	59 093.68	274.57
09	Lena (Color) G	25 886.37	246.99
	B	88 525.34	242.37

7.2. Information entropy

Information entropy serves as a crucial metric, quantifying the degree of uncertainty or unpredictability inherent in an image's information content. Higher values of information entropy signify greater (Li et al., 2018) uncertainty in the image data. For grayscale images with pixel values ranging from 0 to 255, the maximum achievable information entropy is 8. Eq. (6) provides the method to compute information entropy:

$$E(n) = \sum_{i=1}^{2^N-1} P(n_i) \log_2 \left(\frac{1}{P(n_i)} \right) \quad (6)$$

where, N denotes the total number of samples. Each sample, n_i , corresponds to a specific grey level value present in the image. The probability of occurrence for each sample n_i is represented by $P(n_i)$. The information entropy of the ciphered images mentioned earlier was computed, and the outcomes are displayed in Table 6. Furthermore, a comparison was conducted between the suggested scheme and other related schemes in terms of histogram variance and information entropy, as detailed in Table 7. Notably, the results reveal that the information entropy for the suggested scheme demonstrates a stronger correspondence with the theoretical value when compared to the related scheme.

7.3. Correlation coefficient analysis

The correlation coefficient (Cr) is an important parameter for assessing the encryption algorithm's performance. The correlation coefficient between two pixels, x and y , is calculated using Eqs. (7) through (10), yielding a value between -1 and 1 . To withstand statistical attacks effectively, an encryption scheme should strive for correlation coefficients close to zero for adjacent pixels in all three directions: "horizontally (H), vertically (V), and diagonally (D)". Correlation coefficients close to zero indicate a lack of correlation between adjacent pixels, whereas Coefficients near -1 and 1 indicate "strong negative and positive correlations, respectively" (Farah et al., 2020). An encryption scheme exhibiting non-zero correlations become susceptible to statistical attacks, thereby compromising its security.

$$Cr_{xy} = \frac{cov_{xy}}{\sqrt{D_x} \sqrt{D_y}} \quad (7)$$

$$cov_{xy} = \frac{1}{N} \sum_{i=1}^N (x_i - E_x)(y_i - E_y) \quad (8)$$

$$D_x = \frac{1}{N} \sum_{i=1}^N (x_i - E_x)^2 \quad (9)$$

$$E_x = \frac{1}{N} \sum_{i=1}^N (x_i) \quad (10)$$

To evaluate the proposed scheme, a random selection of 4000-pixel pairs from both plain and encrypted images was performed. The

correlation coefficient of each pair was then calculated using the provided equations. Table 4 presents compelling evidence of a significant correlation between adjacent pixels in the plain image. In stark contrast, Table 8 clearly illustrates the absence of any correlation among adjacent pixels in the encrypted image, horizontally (H), vertically (V), and diagonally (D).

These disparities between Tables 4 and 8 are visually evident in the sub-figures depicted in Fig. 12. The first six sub-figures exhibit strong pixel correlations, while the last six sub-figures demonstrate a lack of correlation. Furthermore, the computed correlation coefficient values presented in Table 9 is consistent with those observed in other encryption schemes documented in the literature.

7.4. Key space analysis

The key space is a critical aspect of any cryptosystem, directly impacting the security of the encryption algorithm. A larger key space is essential for a robust encryption algorithm to withstand exhaustive attacks. According to Alvarez and Li (2006), an image encryption method must have a key space greater than 2^{100} in order to effectively withstand brute force assaults. In our approach, we use a secret key $k = (w_0, x_0, y_0, z_0, a_0, b_0, c_0, d_0)$, with eight variables, and since the initial value is a double precision number with a calculation precision of 10^{-16} , the total key space exceeds 2^{425} , which is more than sufficient to withstand all types of brute-force attacks. Table 10 presents the key space analysis of our proposed image encryption approach compared to other existing methods. To retrieve the original image, the adversary must successfully predict all eight elements of the shared secret key and additionally determine the hash value of the pixel data in the plain image.

7.5. Differential attack analysis

Attackers attempt to identify differences in the ciphertext caused by different plaintext pictures in order to undermine an image encryption scheme. This approach is known as a differential or selective plaintext assault. In order to thwart differential attacks or selective plaintext attacks, it is crucial for an image encryption algorithm to exhibit high sensitivity to alterations in the original image. The ciphertext generated by the algorithm must exhibit significant variations when slight changes are made to the plaintext image. This sensitivity prevents attackers from detecting changes easily and ensures strong defence against such attacks. To assess the resistance of an image encryption algorithm to differential attacks, two effective metrics are employed: "the Number of Pixel Changing Rate (NPCR) and the Unified Average Change Intensity (UACI)" (Preishuber et al., 2018). These metrics play a crucial role in assessing the algorithm's capability to withstand potential differential attacks and contribute to enhancing its overall security. The mathematical expression for NPCR and UACI are defined as follows:

$$NPCR = \frac{\sum_{i=1}^M \sum_{j=1}^N C_{ij}}{M \times N} \quad (11)$$

$$UACI = \frac{\sum_{i=1}^M \sum_{j=1}^N |I_1(i, j) - I_2(i, j)|}{M \times N \times 255} \quad (12)$$

where,

$$C_{ij} = \begin{cases} 0, & I_1(i, j) = I_2(i, j) \\ 1, & I_1(i, j) \neq I_2(i, j) \end{cases}$$

Here, $I_1(i, j)$ and $I_2(i, j)$ represent the encrypted images before and after modifying one pixel of the original plaintext image of dimension $M \times N$, respectively. Both encrypted images I_1 and I_2 are encrypted using the same key to yield two sets of encrypted results. NPCR and UACI scores have been computed for every encryption outcome and are presented in Table 11. A recent demonstration showcased that for

Table 6
Information entropy of ciphered images.

Image	Lena256	Plant	Clock	Lena512	Couple	Boat	Chat X-ray	Brain MRI	Lena color		
									Red	Green	Blue
Entropy	7.9974	7.9973	7.9972	7.9994	7.993	7.9993	7.9971	7.9972	7.9973	7.9967	7.9976

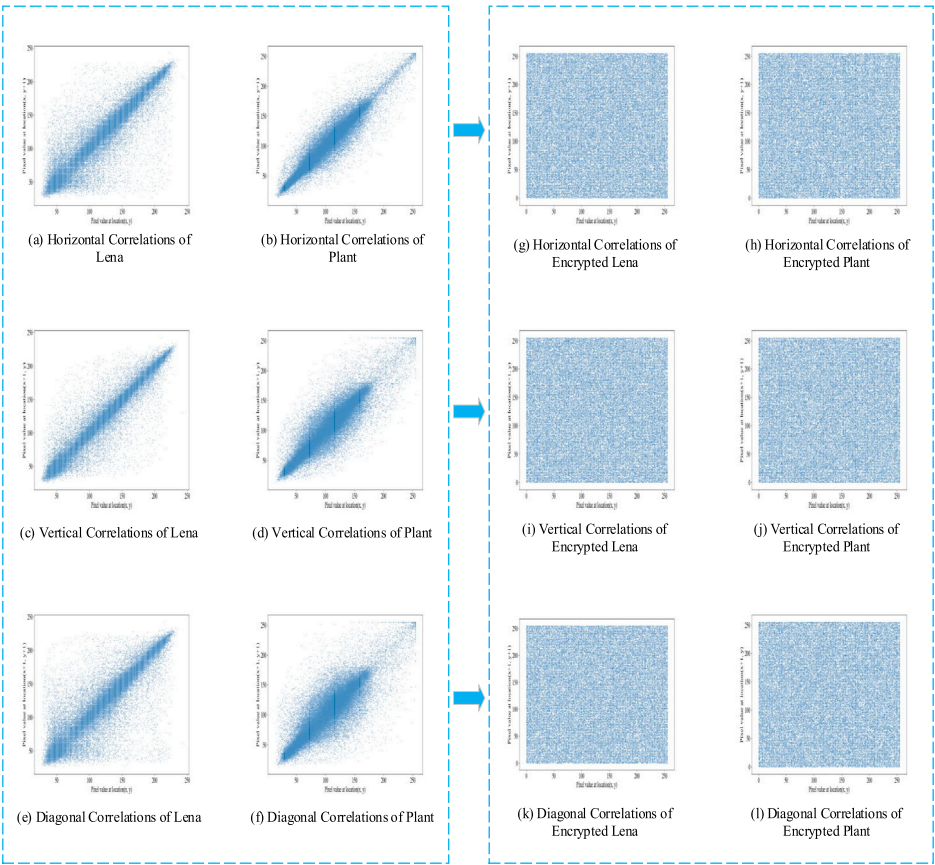


Fig. 12. Visual depiction illustrating the correlation among neighboring pixel values in horizontal, vertical, and diagonal orientations for both original and encrypted images.

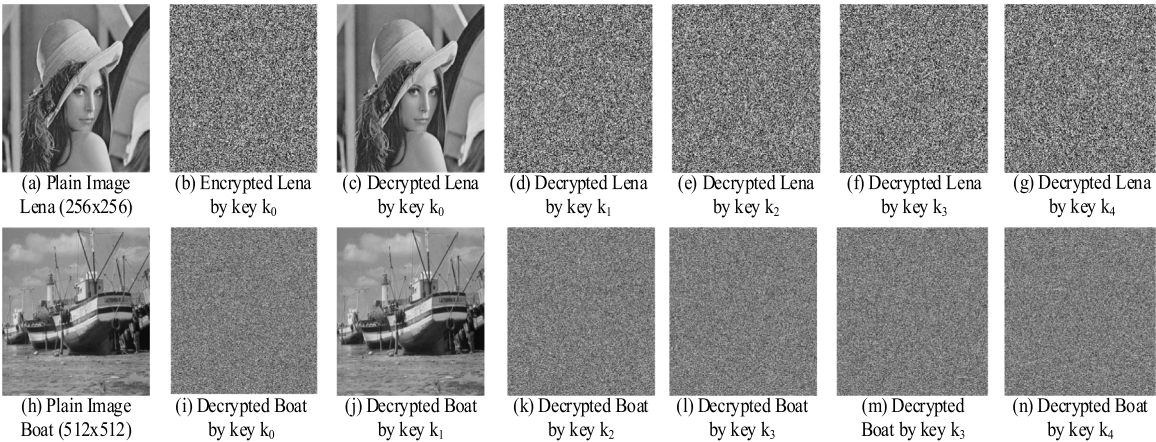


Fig. 13. Key sensitivity analysis.

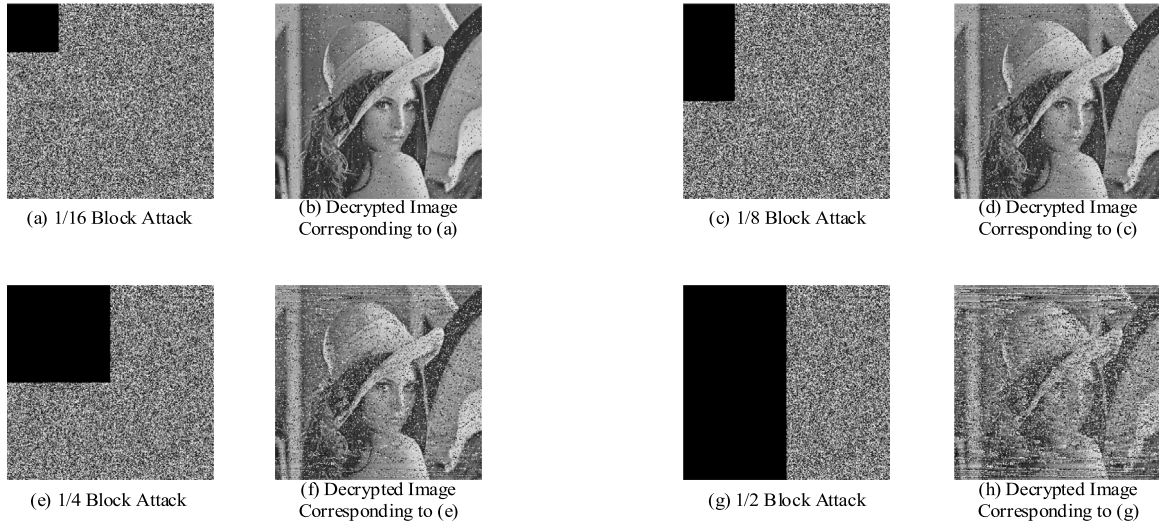


Fig. 14. Image encryption/decryption effects: ((i)–(ix)) Plain images, ((x)–(xviii)) Corresponding encrypted images, and ((xix)–(xxvii)) Corresponding decrypted images.

Table 7

Comparison of histogram variance and information entropy with existing schemes.

Approaches	Histogram variance	Entropy
Our	244.66	7.9974
Zheng and Bao (2023)	245.26	7.9972
Zhao et al. (2023)	270.39	7.9969
Maniyath and V (2020)	–	7.9854
Maazouz et al. (2022)	–	7.9971
Roy et al. (2021)	–	7.9879
Zhang (2020)	–	7.9967
Rani et al. (2023)	–	7.9972

Table 8

Correlation coefficient analysis.

S. No	Image	Horizontal	Vertical	Daigonal
01	Lena 256	–0.00061	–0.00051	0.00034
02	Plant	–0.00121	0.00424	–0.00131
03	Clock	–0.00628	0.00776	0.00435
04	Lena 512	–0.00127	–0.00208	0.00351
05	Couple	–0.00011	–0.00053	–0.00069
06	Boat	–0.00037	0.00094	–0.00055
07	Chest Xray	0.00217	–0.00149	0.00052
08	Brain MRI	0.00723	0.00598	–0.00367
	R	0.00883	0.00630	–0.00548
09	Lena (Color) G	0.00071	–0.00432	0.00294
	B	–0.00339	–0.00101	0.00197

countering differential attacks on an 8-bit grayscale test image, the anticipated NPCR and UACI scores stand at approximately ‘99.6094% and 33.4635%, respectively’ (Midoun et al., 2021). In Table 12, the NPCR and UACI scores of propose encryption scheme with five other image encryption methods are presented. Among these schemes, SSTM-IES demonstrates scores of NPCR and UACI that are closest to the expected values.

7.6. Key sensitivity analysis

The image encryption method must demonstrate a high sensitivity to its secret key, where even a slight change of a single bit in the key leads to a completely different encrypted output (Liu et al., 2019). The paper employs distinct keys to encrypt nine standard images and quantifies the disparities using NPCR and UACI metrics. Table 13 outlines the notable distinctions observed in two ciphered images as the security key transitions from ‘ k ’ to $k' = k + 10^{-15}$. Furthermore, a comparative investigation involving various existing approaches was

undertaken to assess the key sensitivity using NPCR and UACI metrics. The results of this analysis are presented in Table 14, and it is evident that our proposed scheme demonstrates highly favorable outcomes in comparison.

Another aspect of the key sensitivity analysis involves examining the decrypted images obtained with the four one-bit altered keys, using the ciphertext encrypted with the original key. Fig. 13 illustrates the deciphered images. As expected, the image decrypted with the original key (Key 1) matches the original plaintext image (Fig. 13(b) & (j)). However, the other deciphered images are significantly different from the original plaintext image. This finding confirms the high sensitivity of our algorithm to variations in the secret key.

7.7. MSE and PSNR analysis

The Mean Squared Error (MSE) and Peak Signal to Noise Ratio (PSNR) are commonly utilized metrics to assess the accuracy and reliability for any encryption algorithm by comparing the original and encrypted images. MSE quantifies the average squared difference between the pixel values of the two images, while PSNR measures the changes in pixel values between the plain text image and its corresponding encrypted image (Alexan et al., 2023; Mondal & Singh, 2022). These metrics are calculated using the following equations:

$$MSE = \frac{\sum_{i=1}^M \sum_{j=1}^N (I_{(i,j)} - EI_{(i,j)})^2}{M \times N} \quad (13)$$

$$PSNR = 10 \log \left(\frac{I_{max}^2}{MSE} \right) \quad (14)$$

where $M \times N$ is the resolution of the image, $I_{(i,j)}$ denotes the pixel value of the original plain image, and $EI_{(i,j)}$ represents the pixel value of the ciphered image. High MSE and low PSNR values indicate significant differences between the ciphered image and its corresponding original image. Table 15 displays the outcomes of MSE and PSNR obtained through our proposed encryption algorithm across different images. Additionally, Table 16 offers a comparative analysis with existing literature regarding both MSE and PSNR values. The results demonstrate that our proposed algorithm outperforms (Arif et al., 2022) in terms of MSE and PSNR values, exhibits comparable performance to others.

7.8. SSIM

The Structural Similarity Index Measure (SSIM) evaluates the resemblance between the plaintext image and the encrypted image by examining the following aspects: the brightness, contrast, and structure. The

Table 9
Comparison of correlation coefficients with related schemes.

	Horizontal	Vertical	Diagonal
Our	-6.11×10^{-4}	-5.12×10^{-4}	3.46×10^{-4}
Zheng and Bao (2023)	1.8×10^{-3}	-2×10^{-3}	0.1×10^{-3}
Zhao et al. (2023)	6.7981×10^{-4}	2.7856×10^{-4}	3.3×10^{-3}
Zhang (2020)	-2.23×10^{-2}	-8.4×10^{-3}	-8.6×10^{-3}
Maazouz et al. (2022)	3.0×10^{-4}	-3.7×10^{-3}	2.0×10^{-3}
Maniyath and V (2020)	-6.4×10^{-3}	2.2×10^{-2}	1.6×10^{-2}
Rani et al. (2023)	1.4×10^{-2}	2.7×10^{-2}	0.6×10^{-2}

Table 10
Key space analysis.

Algorithm	Key space
Our	2^{425}
Zheng and Bao (2023)	$10^{48} \approx 2^{159}$
Maazouz et al. (2022)	2^{356}
ur Rehman et al. (2018)	2^{375}
Mansoor and Parah (2023)	2^{215}
Muñoz-Guillermo (2021)	2^{448}
Abduljabbar et al. (2022)	$10^{64} \approx 2^{212}$
Rani et al. (2023)	2^{215}

SSIM value ranges from 0 to 1, with higher values suggesting a higher degree of similarity between the encrypted and plain images (El-Shafai et al., 2021). The SSIM formula is precisely defined as follows:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (15)$$

where:

μ_x = average of x
 μ_y = average of y
 σ_x^2 = variance of x
 σ_y^2 = variance of y
 σ_{xy} = covariance of x and y
 $C_1 = (K_1 L)^2$
 $C_2 = (K_2 L)^2$

with $K_1 = 0.01$ and $K_2 = 0.03$ being small constants, and L being the dynamic range of the pixel values.

Table 17 demonstrates that the SSIM values of the unaltered photos and their equivalent encrypted images are close to 0, suggesting a significant dissimilarity between them. Furthermore, the comparison conducted in Table 18 demonstrates that the proposed SSTM-IES outperforms current Chaos based image encryption systems (C-IES) in terms of SSIM values. These findings suggest that the suggested image encryption model is very efficient and resistant to statistical attacks, highlighting its appropriateness for practical applications.

7.9. Occlusion attack

During the transmission of data from the sender to the receiver, there is a potential for the encrypted data to undergo distortion due to an insecure channel, which could lead to image loss; therefore, an effective image encryption scheme should be resilient against such attacks (El Ogr et al., 2023). In order to analyze the susceptibility to cropping attacks, we cropped a section of pixels from the encrypted image, as illustrated in Fig. 14, and then applied the inverse process of the proposed encryption scheme to the cropped cipher images. The resulting decrypted image exhibits a high resemblance to the original plaintext image. For a quantitative assessment, we computed the PSNR and MSE metrics, which are shown in Table 19. We then conducted a comparison of the average PSNR between our proposed method and a reference method, as indicated in Table 20.

8. Applying SSTM-IES in video encryption: Analysis and findings

This section introduces a strong video encryption solution that incorporates the SSTM-IES encryption algorithm from Section 5. This integration is an important advancement in strengthening the security measures for digital information and is a critical step towards protecting video data. A thorough illustration of the suggested video encryption technique is shown in Fig. 15. In this method, the video is handled like a collection of frames. Encryption begins with frame extraction, which interprets each frame as a digital image. Two of these frames, frames 1 and 2, are selected to be vital frames and are discovered using Algorithm 6. Both key frames are encrypted using the SSTM-IES method, resulting in encrypted key frames F1 and F2. This technique further processed by categorizing frames into odd and even indexes, and then performs XOR operations between the frames and the encrypted key frames. Odd indexed frames interact with encrypted key frame F1, while even indexed frames interact with encrypted key frame F2. Together, these encrypted frames create the final encrypted video.

Algorithm 6

Input: Key($w_0, x_0, y_0, z_0, a_0, b_0, c_0, d_0$)
Output: Frame index numbers ($F1, F2$)
 $M \leftarrow w_0 + x_0 + y_0 + z_0$
 $N \leftarrow a_0 + b_0 + c_0 + d_0$
 $F1 \leftarrow \lfloor M \times 10^3 \rfloor \bmod 31$
 $F2 \leftarrow \lfloor N \times 10^3 \rfloor \bmod 31$
If $F1 = F2$ **then**
 $F2 \leftarrow F1 + 30$
Return $F1, F2$

This method's effectiveness and security were assessed by applying it to common video samples, such as the Foreman video from the derf collection (media.xipn.org). A frame rate of 30 frames per second (fps) (Wang et al., 2017) was used throughout the studies utilizing this video encryption approach. A few brief security evaluations of the suggested encryption method have been carried out in order to determine its resistance to several types of attacks, such as brute force, differential, and statistical types of attacks. The results gained from this study are presented in Table 21 for several frames. Furthermore, this method is compared with other video encryption schemes in a comparative analysis, which is shown in Table 22.

The thorough review of the data and the analysis that followed confirm that the recommended technique for video encryption works and satisfies the strict requirements needed for safe encryption techniques.

9. Discussion

In recent literature, a variety of chaotic map-based image encryption schemes have been introduced, each of which has demonstrated substantial success in the areas of randomness, security, and statistical analysis. However, a significant number of these schemes offer minimal or nonexistent comprehensive security assessments. Conducting comprehensive security evaluations is crucial due to the wide use of chaos-based image encryption methods (C-IES).

Table 11
NPCR and UACI analysis.

Image	Lena256	Plant	Clock	Lena512	Couple	Boat	Chest X-ray	Brain MRI	Lena color
NPCR	99.6087	99.6020	99.6351	99.6006	99.6057	99.6092	99.5911	99.6019	99.5963
UACI	33.4459	33.4649	33.3493	33.4295	33.4198	33.4275	33.3951	33.4562	33.4307

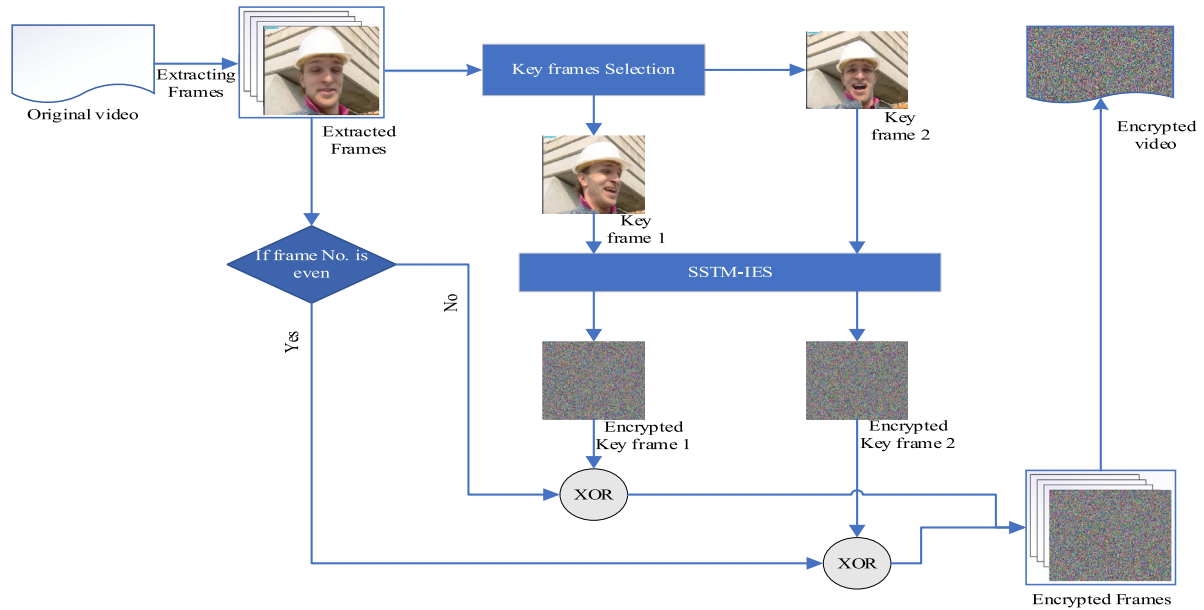


Fig. 15. A schematic representation of the proposed video encryption scheme.

Table 12
Comparison of NPCR and UACI with other schemes.

	UACI	NPCR
Theoretical value	33.4635	99.6094
Our	33.4459	99.6087
Zheng and Bao (2023)	33.51	99.62
Zhao et al. (2023)	33.4789	99.5935
Zhang (2020)	33.4636	99.6114
Maazouz et al. (2022)	33.27	99.59
Maniyath and V (2020)	33.61	99.87
Rani et al. (2023)	33.47	99.60

Table 13
NPCR and UACI values in the key sensitivity analysis.

S. No.	Image	UACI	NPCR
01	Lena256	33.5060	99.5999
02	Plant	33.5068	99.6099
03	Clock	33.4528	99.5946
04	Lena512	33.4879	99.6040
05	Couple	33.4738	99.6088
06	Boat	33.5056	99.6080
07	Chest X-ray	33.4941	99.6011
08	Brain MRI	33.5062	99.6012
09	Lena color	33.4507	99.6063

Table 23 displays a comprehensive evaluation of the proposed C-IES in comparison to existing schemes. The evaluation is based on various security analysis criteria and attributes. The table uses “✓” to signify the existence of a specific security feature and “✗” to indicate its nonexistence. The final row provides a summary of the scores for each C-IES, based on the amount of attributes that were assessed.

The results demonstrate that proposed SSTM-IES has undergone a comprehensive security evaluation and has received the highest scores when compared to other schemes. Furthermore, the suggested SSTM-IES has proven its ability to be used for video encryption, hence solidifying its superiority compared to other encryption methods.

The proposed SSTM-IES encryption system demonstrates exceptional performance in multiple measures. Table 9 exhibits the minimum cross-correlation value, which signifies a high level of encryption quality. Table 7 has an information entropy score of 7.9974, which is very near to the ideal value of 8.0. This suggests a higher level of randomness. Table 12 demonstrates the high-quality performance of the system, as indicated by the near-perfect NPCR and UACI values. This indicates that the system is highly resistant to differential attacks. Table 20 exhibits the ability to withstand occlusion attacks, as evidenced by a PSNR score of 12.29, even when half of the image is lost. This ensures that the image remains comprehensible. This thorough assessment highlights the SSTM-IES’s resilience and adaptability and shows off its potential for a range of high-security applications.

9.1. Implication and limitations

This section discusses the implications and the limitations of the proposed work.

9.1.1. Implication

The suggested image encryption technique has numerous practical implications, namely for sectors that manage sensitive multimedia information, such as healthcare, telemedicine, education, and biometric verification. The technology strengthens data security and patient confidentiality by using strong encryption for medical pictures and videos, which is essential for meeting regulatory requirements. The method’s outstanding effectiveness and minimal computing burden make it well-suited for real-time applications in cloud environments. This enables secure transmission and storage of data without any noticeable decrease in performance. Furthermore, the technique’s capacity to be easily adjusted for use in other areas, such as analyzing microbiological images, highlights its versatility and wide range of applications in scientific and medical research. This ensures that sensitive data remains secure and confidential.

Table 14

Comparison of NPCR and UACI for the key sensitivity test with other schemes.

	Theoretical value	Our	Zhang (2020)	Abduljabbar et al. (2022)	Yousif et al. (2020)
NPCR	99.6094	99.5999	99.6095	99.6002	99.95
UACI	33.4635	33.5060	33.4705	30.3924	46.82

Table 15

MSE and PSNR values between plaintext and cipher image.

Image	Lena256	Plant	Clock	Lena512	Couple	Boat	Chest X-ray	Brain MRI	Lena color
MSE	7769.35	7791.31	12174.09	7766.86	7084.08	7652.40	16301.65	16296.60	8919.59
PSNR	9.2	9.2	7.2	9.2	9.6	9.2	6.0	6.0	8.6

Table 16

Comparison of MSE and PSNR values with related encryption scheme.

	Our	Mansoor and Parah (2023)	Arif et al. (2022)	Roy et al. (2021)	Yousif et al. (2020)
MSE	7769.35	–	–	7179.65	7772.2
PSNR	9.22	9.2736	9.23763	9.57	9.2254

Table 17

SSIM values between plaintext and cipher images.

Image	SSIM
Lena256	0.00083
Plant	0.00071
Clock	0.0010
Lena512	0.00097
Chest X-ray	0.00086
Brain MRI	0.00081

Table 18

Comparison of SSIM values with other related schemes.

Approaches	SSIM
Theoretical value	0.0
Our	0.0013
Zhang (2020)	0.0080
Abduljabbar et al. (2022)	0.0021
Yousif et al. (2020)	0.0095

9.1.2. Limitations

The suggested encryption method improves cloud environment security measures for multimedia data in an efficient manner. Nevertheless, there are certain current constraints that must be resolved in subsequent studies:

- The current investigation largely concentrates on standard grayscale and single-color images with uniform resolutions, encompassing two medical images. It is imperative that real-world photos with different resolutions be analyzed in order to increase the encryption scheme's application. Real-world images are usually more diverse and have greater resolutions, which might increase the system's overall applicability and produce different results.

- The video encryption analysis encompassed many videos, which were assessed at a standardized frame rate of 30 frames per second (fps). As different rates may yield different results, it is advised to test with other widely used frame rates. Moreover, although multiple statistical tests were performed on the video encryption, additional experimental assessments are required to verify its appropriateness for real-world applications. The suggested encryption technique provides efficient security measures for safeguarding multimedia data in cloud environments. However, it is important to acknowledge and resolve many existing limitations:

10. Conclusion and future work

This study addresses the security challenges prevalent in cloud storage environments, which are particularly vulnerable due to their open and multi-tenant nature. Ensuring the integrity of security measures is

of utmost importance when establishing a secure repository for sensitive image data in the cloud. This research introduces an innovative and robust image encryption approach tailored specifically for cloud storage systems. The proposed methodology introduces a novel modified chaotic map known as SSTM, which exhibits strong and effective chaotic behavior. SSTM demonstrates key qualities including unpredictability and high sensitivity to initial conditions. These attributes are validated through dynamic analysis methods like bifurcation diagrams and Lyapunov exponents. The complexity of the suggested map is underscored by its notably high Lyapunov exponent value ($LE = 15.98$).

The encryption scheme comprises two core components: a confusion architecture based on the two rounds of permutation method followed by diffusion. This combined approach enhances resilience against various attack vectors. The integration of the two rounds of permutation based shuffling introduces an added layer of robustness and complexity to the overall scheme. The SSTM-IES algorithm, as proposed, emerges as a promising solution that surpasses existing algorithms across multiple performance parameters. With a substantial key space of 2^{425} , this method offers significantly enhanced security. Comprehensive experimental analyses confirm the scheme's robustness against a range of attacks, including differential attacks and entropy analysis. Quantitative assessments demonstrate impressive average NPCR and UACI values of 99.6087% and 33.4459%, respectively. Looking forward, there is great potential to extend the application of chaos based security solutions to encompass multimedia data, including audio and video, stored within cloud environments. This opens up exciting avenues for future research in the field of securing diverse forms of digital assets. Additionally, the article presents a video encryption scheme, demonstrating its robustness through comprehensive results and analysis. Looking ahead, there exists significant potential to expand chaos-based security solutions to encompass diverse multimedia data, such as audio and real time video, processed within cloud environments. This paves the way for intriguing avenues in future research dedicated to securing a wide array of digital assets.

In the future, there is a great opportunity to extend chaos-based security solutions to include various types of multimedia data, such as audio and real-time video, in cloud systems. The proposed encryption approach can bolster the security of microbiological image analysis by safeguarding sensitive data, upholding picture integrity, and thwarting unauthorized access (Pan et al., 2022). The SSTM-IES algorithm efficiently manages intricate pictures, maintaining their integrity during the process of encryption and decryption, which is essential for precise microbiological research. Although the assessment has been performed on standard images, it is necessary to evaluate it using real-time data, which frequently consists of high-resolution and varied types of images, in order to maintain consistency and highlight areas that need improvement. Furthermore, doing a thorough examination of video encryption,

Table 19

Values of MSE and PSNR between recovered images from cropped cipher image and original image.

Image	1/16		1/8		1/4		1/2	
	MSE	PSNR	MSE	PSNR	MSE	PSNR	MSE	PSNR
Lena256	474.67	21.3668	956.89	18.3221	1911.70	15.3165	3835.41	12.2926
boat512	476.49	21.3501	954.67	18.3322	1895.54	15.3534	3800.19	12.3327

Table 20

Comparison of PSNR and MSE values after cropping attack with other existing schemes.

Image	1/16		1/8		1/4		1/2	
	MSE	PSNR	MSE	PSNR	MSE	PSNR	MSE	PSNR
Our	474.67	21.3668	956.89	18.3221	1911.70	15.3165	3835.41	12.2926
Abduljabbar et al. (2022)	534.36	20.8524	1080.91	17.7929	2225.15	14.6572	5126.84	11.0323
Qian et al. (2021)	540.18	20.8054	1758.00	15.6806	2161.82	14.7826	10828.80	7.7850
Yang et al. (2022)	539.87	20.8079	–	–	2150.80	14.8048	4316.08	11.7799
Yousif et al. (2020)	–	–	1265.26	17.1090	4401.90	11.6944	8804.81	08.6836
Roy et al. (2021)	1130.00	17.60	1640.89	15.98	2438.26	14.26	4592.82	11.51

Table 21

Analysis of video encryption.

		Channel	Frame_1	Frame_2	Frame_3	Average
Histogram analysis						
Original video sequence		R	73 759.05	71 640.10	71 858.16	72,419.10
		G	63 446.89	59 386.11	60 304.90	61,045.96
		B	40 211.75	39 443.93	39 733.96	39,796.54
Encrypted video sequence		R	236.40	239.46	254.92	243.59
		G	268.00	252.47	289.05	269.84
		B	254.17	257.43	254.92	255.50
Correlation analysis						
Original video sequence	Horizontal	R	0.9714	0.9710	0.9695	0.9706
		G	0.9776	0.9773	0.9765	0.9771
		B	0.9779	0.9778	0.9772	0.9776
	Vertical	R	0.9702	0.9692	0.9695	0.9696
		G	0.9766	0.9762	0.9764	0.9764
		B	0.9774	0.9771	0.9775	0.9773
	Diagonal	R	0.9543	0.9535	0.9528	0.9535
		G	0.9643	0.9638	0.9637	0.9639
		B	0.9648	0.9650	0.9651	0.9649
Encrypted video sequence	Horizontal	R	0.0536	0.00698	0.00619	0.00617
		G	0.0844	0.00988	0.00904	0.00912
		B	0.0495	0.00723	0.00558	0.00592
	Vertical	R	0.00774	0.00673	0.00785	0.007446
		G	0.00148	0.00267	0.00172	0.00195
		B	0.00704	0.00484	0.00646	0.00611
	Diagonal	R	0.00663	0.00637	0.00719	0.00673
		G	0.00973	0.00629	0.00921	0.00841
		B	0.00638	0.00656	0.00558	0.00617
NPCR & UACI analysis between original and encrypted video						
Original vs. Encrypted video sequence	NPCR	99.62	99.59	99.60	99.60	
	UACI	29.11	29.16	29.05	33.11	
NPCR & UACI analysis between encrypted videos with different keys						
Encrypted video sequence (Encrypted via two slightly different keys)	NPCR	99.58	99.62	99.59	99.59	
	UACI	32.59	33.10	33.27	32.98	

Table 22

Comparison of video encryption approach analysis with existing schemes.

Approaches	Histogram variance	Correlation analysis			NPCR & UACI analysis	
		Horizontal	Vertical	Diagonal	NPCR	UACI
Our	243.59	0.00592	0.00195	0.00617	99.60	33.11
Elkamchouchi et al. (2020)	NA	0.0025	0.0037	0.0049	98.6327	34.0338
Dhingra and Dua (2024)	NA	−0.0012	−0.0002	−0.0006	99.65	50.02
Sallam et al. (2018)	NA	−0.2533	−0.2557	−0.2126	NA	NA
El Ogri et al. (2023)	292.54	0.0073	0.0052	0.0064	99.98	33.80

Table 23

Analysis of suggested SSTM-IES in comparison to the existing C-IES schemes.

Scheme	Histogram analysis	Information entropy	Local entropy	Cross correlation	SSIM	MSE & PSNR	NPCR	UACI	Occlusion attack	Video applicability	Key space	Key sensitivity	Comparison	Total
Hu et al. (2020)	✓	✓	✓	✓	✗	✗	✓	✓	✓	✗	✓	✓	✗	9/13
Kumar et al. (2019)	✗	✓	✗	✓	✓	✗	✓	✓	✓	✗	✓	✓	✗	8/13
Mondal et al. (2019)	✓	✓	✗	✓	✗	✓	✓	✓	✓	✗	✓	✓	✗	9/13
Zhang et al. (2020)	✓	✓	✗	✓	✗	✗	✓	✓	✓	✗	✓	✓	✗	8/13
Falmari and Brindha (2020)	✓	✓	✗	✗	✗	✗	✓	✓	✗	✗	✓	✓	✗	6/13
Maniyath and V (2020)	✗	✓	✗	✓	✓	✓	✓	✓	✗	✗	✗	✗	✗	6/13
Roy et al. (2021)	✓	✓	✗	✓	✗	✓	✓	✓	✗	✗	✓	✓	✓	9/13
Ahmad and Shin (2021)	✓	✓	✗	✓	✗	✓	✓	✓	✓	✗	✓	✓	✗	8/13
Li, Yu, Wang (2021)	✓	✓	✗	✓	✓	✗	✓	✓	✓	✗	✓	✓	✗	10/13
Maazouz et al. (2022)	✓	✓	✗	✓	✗	✗	✓	✓	✗	✗	✗	✓	✗	6/13
Zhang and Zhang (2022)	✓	✓	✗	✓	✓	✗	✓	✓	✓	✗	✓	✗	✗	6/13
Mondal and Singh (2022)	✓	✓	✗	✓	✗	✓	✓	✓	✗	✗	✗	✗	✗	7/13
Alexan et al. (2023)	✓	✓	✗	✓	✗	✗	✓	✓	✗	✗	✓	✓	✗	7/13
Rani et al. (2023)	✓	✓	✗	✓	✓	✗	✓	✓	✓	✗	✓	✓	✗	9/13
Alawida (2024)	✓	✗	✗	✓	✗	✗	✓	✓	✗	✗	✓	✓	✗	5/13
Shao et al. (2023)	✗	✓	✗	✓	✗	✓	✓	✓	✓	✗	✓	✓	✗	6/13
Our	✓	✓	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	12/13

which includes a larger number of films and greater frame rates, will guarantee the algorithm's effectiveness across different formats. By integrating supplementary multimedia data, the method's versatility in protecting various digital assets will be showcased. Incorporating robust authentication mechanisms will further enhance the security, ensuring that only authorized users can access the encrypted data. Enhancing efficiency and performance in cloud environments can be achieved by optimizing the encryption process for parallel processing. This approach is particularly useful for large-scale applications.

CRedit authorship contribution statement

Talha Umar: Methodology, Formal analysis, Data curation, Writing – original draft, Writing – review & editing. **Mohammad Nadeem:** Conceptualization, Supervision, Data curation, Writing – review & editing. **Faisal Anwer:** Conceptualization, Supervision, Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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